

# Phenix Maps and Density

13 documentation pages

Generated from local Phenix documentation

## Contents

1. Density Modification
2. Generating a composite omit map
3. Cutting density out of a map with `cut_out_density`
4. Merging two models with `density_outside_model`
5. `phenix.fem`: compute Feature Enhanced Map (FEM)
6. Finding NCS from density in a map with `find_ncs_from_density`
7. `phenix.map_box`: extract box with model and map around selected atoms
8. Map comparison: Identify matching contour levels in two maps
9. Map correlations (map-model, map-map, optional origin shifts)
10. Mapping one PDB file onto another using space-group symmetry with `phenix.map_to_object`
11. `phenix.map_to_structure_factors`: convert map into structure factors
12. `<string>`
13. Creating maps and map coefficients with `phenix.maps`

# Density Modification

---

density-modification.html

## Why

The crystallographic phase problem can be solved using experimental phasing or molecular replacement. In both cases, these initial phases may be inaccurate due to experimental errors, low signal-to-noise, or model bias. Therefore, it is usually necessary to improve the phases by exploiting our prior knowledge about electron density distributions in the crystal. In the case of single wavelength (SAD) or single derivative (SIR) data, this step is essential for resolving the inherent phase ambiguity in these experiments.

The phase improvement procedure is usually referred to as density modification because new phase estimates are obtained by modifying the electron density to conform to prior expectations. For example, methods to improve phases include solvent flattening, histogram matching, and non-crystallographic symmetry averaging (if NCS is present). The result of the procedure should be a better set of phases, which can be used to make a map for interpretation (i.e., model building).

## How

In Phenix, several programs carry out density modification (see the list of Related programs below). Density modification can be performed using either a set of experimental phases (i.e., Hendrickson-Lattmann coefficients) or experimental amplitudes and an atomic model (from which the phases are calculated). The density modification program will try to automatically determine solvent content and non-crystallographic relationships between regions of density.

How to use the phenix Resolve density modification GUI: [Click here](#)

Tutorial on density modification : [Click here](#)

## Related programs

- `phenix.density_modification`: This program performs iterative phase improvement with Resolve density modification, including the use of non-crystallographic symmetry and electron-density distributions.
- `phenix.multi_crystal_average`: This program improves phases interactively and averages electron density, both within a crystal and between crystals. This very powerful technique can generate high quality phases in some circumstances.
- `phenix.ncs_average`: This program averages the electron density for molecules related by non-crystallographic symmetry. No iterative phase improvement is performed, just a single round of averaging. This is useful for improving signal-to-noise in maps for model building.

# Generating a composite omit map

---

composite\_omit\_map.html

## Contents

- Overview
- Running the program
- Single-omit mode
- Other options
- References
- List of all available keywords

## Overview

A composite omit map is an omit map covering the entire contents of the unit cell, generated by computing a number of omit maps covering specific regions, and combining the areas around the omitted regions. This allows a relatively bias-free map to be calculated for the structure without severely compromising phase quality. Because this is significantly slower than computing a simple omit map, it is not recommended if you are only interested in a specific ligand or problematic region of the structure. However, because the procedure is inherently parallel, it may be spread out over multiple processors or cluster nodes to limit the runtime.

The program `phenix.composite_omit_map` is used to generate these maps; internally, it is essentially no more than a wrapper for `phenix.refine` and the underlying map-calculation code. It may be run in three different modes:

with simulated annealing, which aggressively removes residual bias in the partial models (but increases runtime). with minimization only, without annealing without refinement, which leaves the partial models alone, but performs an iterative procedure to quickly remove phase bias.

- with simulated annealing, which aggressively removes residual bias in the partial models (but increases runtime).
- with minimization only, without annealing
- without refinement, which leaves the partial models alone, but performs an iterative procedure to quickly remove phase bias.

The final option is the default, since it is very fast but still effective. For best results we recommend running with simulated annealing, although this may be more problematic for especially poor and/or low-resolution structures.

## Running the program

Execution of the program is very similar to running `phenix.refine`:

```
% phenix.composite_omit_map model.pdb data.mtz ligands.cif nproc=64
```

The requirements for input files and formats are the same. Because the program is intended to be as automatic as possible, only a limited subset of refinement parameters are exposed to the user. Refinement may be disabled by adding `refine=False` to the command line; annealing is switched on by adding `anneal=True`. Like `phenix.refine`, experimental phases (Hendrickson-Lattman coefficients) will be used automatically if present, and you may specify a twin law if necessary.

You may request any supported combination of maps using the same map type strings as elsewhere in Phenix, for instance 2mFo-DFc, anomalous, 3mFo-2DFc, and so on. The default map type is 2mFo-DFc, but the `map_type` keyword may specify one or more alternate maps. You may also change the choice of whether to fill in missing observations with F(calc) (off by default), or whether to exclude reflections flagged for calculating R-free (on by default). NCS averaging is also available as an option.

If simulated annealing is to be run, you may specify the dynamics type (`annealing_type` keyword, either "cartesian" or "torsion") and the starting temperature (`annealing_temperature`, which defaults to 5000K for Cartesian and 2500K for torsion; lower will run more quickly). The atoms left out of structure factor calculation will by default be restrained to their starting positions to prevent the other atoms from filling in the missing scattering. Note that the default behavior allows the bulk solvent mask to extend into the omitted region; this may be turned off by adding the argument `exclude_bulk_solvent=True`, but it is unclear what the best practice is.

The program will parallelize over multi-core systems or managed clusters (including Sun Grid Engine, PBS, and LSF queuing systems) if the `nproc` keyword is given. Use the `technology` keyword to specify the system type. The default behavior is to omit up to approximately 5% of the structure at a time (controlled by the `fraction_omit` keyword), but in practice because of the way the structure is divided there will be significantly more than 20 omit regions for the typical macromolecular structure. If you are unsure how many processes to use, specify `nproc=Auto`.

Output consists of a single MTZ file, named `composite_omit_map.mtz` by default; you may specify a preferred file name using the command line keyword `output.file_name`. This will contain amplitudes and phases for each map type. For 2mFo-DFc maps, the MTZ column labels will be interpretable using the auto-open function in Coot; all other map types require you to load them manually.

## Singe-omit mode

The program can also be used to generate a simple omit map for a single region; the iterative de-bias procedure is not available in this mode, but you can perform refinement and/or simulated annealing. This is essentially equivalent to setting up a `phenix.refine` run manually, but with a simplified interface.

## Other options

There are several other programs in Phenix with similar features:

AutoBuild has an older implementation of this method. A more powerful approach is to generate an "iterative-build" omit map (Terwilliger et al. 2008), which rebuilds the structure instead of annealing. `phenix.refine` can be used to manually generate omit maps for a smaller selection of the model by specifying the `omit_selection` keyword. The single-omit mode of the composite omit map program is equivalent to this. `phenix.maps` can generate simple omit maps without refinement.

- AutoBuild has an older implementation of this method. A more powerful approach is to generate an "iterative-build" omit map (Terwilliger et al. 2008), which rebuilds the structure instead of annealing.
- `phenix.refine` can be used to manually generate omit maps for a smaller selection of the model by specifying the `omit_selection` keyword. The single-omit mode of the composite omit map program is equivalent to this.
- `phenix.maps` can generate simple omit maps without refinement.

## References

Model bias in macromolecular crystal structures. A. Hodel, S.H. Kim, and A.T. Brunger. *Acta Cryst.* A48, 851-858 (1992).

- Model bias in macromolecular crystal structures. A. Hodel, S.H. Kim, and A.T. Brunger. *Acta Cryst.* A48, 851-858 (1992).

Iterative-build OMIT maps: map improvement by iterative model building and refinement without model bias. T.C. Terwilliger, R.W. Grosse-Kunstleve, P.V. Afonine, N.W. Moriarty, P.D. Adams, R.J. Read, P.H. Zwart, and L.-W. Hung. *Acta Cryst.* D64, 515-524 (2008).

- Iterative-build OMIT maps: map improvement by iterative model building and refinement without model bias. T.C. Terwilliger, R.W. Grosse-Kunstleve, P.V. Afonine, N.W. Moriarty, P.D. Adams, R.J. Read, P.H. Zwart, and L.-W. Hung. *Acta Cryst.* D64, 515-524 (2008).

Towards automated crystallographic structure refinement with phenix.refine. P.V. Afonine, R.W. Grosse-Kunstleve, N. Echols, J.J. Headd, N.W. Moriarty, M. Mustyakimov, T.C. Terwilliger, A. Urzhumtsev, P.H. Zwart, and P.D. Adams. *Acta Crystallogr D Biol Crystallogr* 68, 352-67 (2012).

- Towards automated crystallographic structure refinement with phenix.refine. P.V. Afonine, R.W. Grosse-Kunstleve, N. Echols, J.J. Headd, N.W. Moriarty, M. Mustyakimov, T.C. Terwilliger, A. Urzhumtsev, P.H. Zwart, and P.D. Adams. *Acta Crystallogr D Biol Crystallogr* 68, 352-67 (2012).

## List of all available keywords

inputsequence = None scattering\_table = wk1995 it1992 \*n\_gaussian neutron electron wavelength = None energy = None use\_experimental\_phases = Auto twin\_law = Auto Enter twin law if known.xray\_dataScope of X-ray data and free-R flagsfile\_name = None labels = None high\_resolution = None low\_resolution = None twin\_law = None outliers\_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliersfrench\_wilson\_scale = True sigma\_fobs\_rejection\_criterion = None sigma\_iobs\_rejection\_criterion = None ignore\_all\_zeros = True force\_anomalous\_flag\_to\_be\_equal\_to = None convert\_to\_non\_anomalous\_if\_ratio\_pairs\_lone\_less\_than\_threshold = 0.5 french\_wilsonmax\_bins = 60 Maximum number of resolution binsmin\_bin\_size = 40 Minimum number of reflections per binr\_free\_flagsfile\_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None required = True Specify if free-r flags are must be present (or else generated)test\_flag\_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore\_r\_free\_flags = False Use all reflections in refinement (work and test)disable\_suitability\_test = False ignore\_pdb\_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max\_free = 2000 lattice\_symmetry\_max\_delta = 5 use\_lattice\_symmetry = True use\_dataman\_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n\_shells = 20 experimental\_phasesfile\_name = None labels = None pdbfile\_name = None Model file(s) name (PDB)monomersfile\_name = None Monomer file(s) name (CIF)mapsmap\_file\_name = None A CCP4-formatted mapd\_min = None Resolution of mapmap\_coefficients\_file\_name = None MTZ file containing mapmap\_coefficients\_label = None Data label for complex map coefficients in MTZ fileoutputfile\_name = composite\_omit\_map.mtz debug = False Show additional debugging output, and save intermediate files.directory = None Output directory for GUI jobsjob\_title = None Job title in PHENIX GUI, not used on command lineomit\_mapmap\_type = None This should be a string representing standard map types, e.g. 2mFo-DFc. You may specify as many of these as you want (each as a separate argument).omit\_type = \*simple refine anneal None Determines the method used to prepare the model used for phasing omit maps. The simple option de-biases the phases by treating F(model) directly in an iterative procedure. The refine and anneal options instead run phenix.refine on the

partial model(s).fraction\_omit = 0.05 Fraction of atoms to omit at a timeresolution\_factor = 0.25 Grid spacing for assembling partial maps.fill\_missing\_f\_obs = False Fill missing F(obs) with F(calc)exclude\_free\_r\_reflections = True Remove reflections flagged for R-freemap\_types = None This should be a space-separated list of strings representing standard map types, e.g. 2mFo-DFc. Identical to the separate map\_type parameter except that map types are combined into a single string.boxingselection = None Selection of atoms to omit in groups. If not specified, the procedure will use the entire structure. Not compatible with omit\_type=simple.single = False Generates a single omit map rather than instead of dividing into boxes and combining individual maps, equivalent to setting fraction\_omit=1.0. The selection must be specified if this option is used.box\_cushion\_radius = 2.5 Additional buffer around omitted region within which to set occupancies to zero.exclude\_bulk\_solvent = False Prevent bulk solvent from extending into omitted regions.optimize\_binning = True Re-group omit regions to reduce the total number of regions while keeping the fraction omitted per region as similar as possible.refinementannealing\_type = \*cartesian torsion annealing\_temperature = Auto Defaults to 5000K for Cartesian,...

- inputsequence = None scattering\_table = wk1995 it1992 \*n\_gaussian neutron electron wavelength = None energy = None use\_experimental\_phases = Auto twin\_law = Auto Enter twin law if known.xray\_dataScope of X-ray data and free-R flagsfile\_name = None labels = None high\_resolution = None low\_resolution = None twin\_law = None outliers\_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliersfrench\_wilson\_scale = True sigma\_fobs\_rejection\_criterion = None sigma\_iobs\_rejection\_criterion = None ignore\_all\_zeros = True force\_anomalous\_flag\_to\_be\_equal\_to = None convert\_to\_non\_anomalous\_if\_ratio\_pairs\_lone\_less\_than\_threshold = 0.5 french\_wilsonmax\_bins = 60 Maximum number of resolution binsmin\_bin\_size = 40 Minimum number of reflections per binr\_free\_flagsfile\_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None required = True Specify if free-r flags are must be present (or else generated)test\_flag\_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore\_r\_free\_flags = False Use all reflections in refinement (work and test)disable\_suitability\_test = False ignore\_pdb\_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max\_free = 2000 lattice\_symmetry\_max\_delta = 5 use\_lattice\_symmetry = True use\_dataman\_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n\_shells = 20 experimental\_phasesfile\_name = None labels = None pdbfile\_name = None Model file(s) name (PDB)monomersfile\_name = None Monomer file(s) name (CIF)mapsmap\_file\_name = None A CCP4-formatted mapd\_min = None Resolution of mapmap\_coefficients\_file\_name = None MTZ file containing mapmap\_coefficients\_label = None Data label for complex map coefficients in MTZ file
- sequence = None
- scattering\_table = wk1995 it1992 \*n\_gaussian neutron electron
- wavelength = None
- energy = None
- use\_experimental\_phases = Auto
- twin\_law = Auto Enter twin law if known.
- xray\_dataScope of X-ray data and free-R flagsfile\_name = None labels = None high\_resolution = None low\_resolution = None twin\_law = None outliers\_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliersfrench\_wilson\_scale = True sigma\_fobs\_rejection\_criterion = None sigma\_iobs\_rejection\_criterion = None ignore\_all\_zeros = True force\_anomalous\_flag\_to\_be\_equal\_to = None

convert\_to\_non\_anomalous\_if\_ratio\_pairs\_lone\_less\_than\_threshold = 0.5 french\_wilsonmax\_bins = 60  
 Maximum number of resolution binsmin\_bin\_size = 40 Minimum number of reflections per  
 binr\_free\_flagsfile\_name = None This is normally the same as the file containing Fobs and is usually  
 selected automatically.label = None required = True Specify if free-r flags are must be present (or else  
 generated)test\_flag\_value = None This value is usually selected automatically - do not change unless  
 you really know what you're doing!ignore\_r\_free\_flags = False Use all reflections in refinement (work and  
 test)disable\_suitability\_test = False ignore\_pdb\_hexdigest = False If True, disables safety check based  
 on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free  
 flags (if not available in input files)fraction = 0.1 max\_free = 2000 lattice\_symmetry\_max\_delta = 5  
 use\_lattice\_symmetry = True use\_dataman\_shells = False Used to avoid biasing of the test set by certain  
 types of non-crystallographic symmetry.n\_shells = 20

- file\_name = None
- labels = None
- high\_resolution = None
- low\_resolution = None
- twin\_law = None
- outliers\_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliers
- french\_wilson\_scale = True
- sigma\_fobs\_rejection\_criterion = None
- sigma\_iobs\_rejection\_criterion = None
- ignore\_all\_zeros = True
- force\_anomalous\_flag\_to\_be\_equal\_to = None
- convert\_to\_non\_anomalous\_if\_ratio\_pairs\_lone\_less\_than\_threshold = 0.5
- french\_wilsonmax\_bins = 60 Maximum number of resolution binsmin\_bin\_size = 40 Minimum number of reflections per bin
- max\_bins = 60 Maximum number of resolution bins
- min\_bin\_size = 40 Minimum number of reflections per bin
- r\_free\_flagsfile\_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None required = True Specify if free-r flags are must be present (or else generated)test\_flag\_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore\_r\_free\_flags = False Use all reflections in refinement (work and test)disable\_suitability\_test = False ignore\_pdb\_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max\_free = 2000 lattice\_symmetry\_max\_delta = 5 use\_lattice\_symmetry = True use\_dataman\_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n\_shells = 20
- file\_name = None This is normally the same as the file containing Fobs and is usually selected automatically.
- label = None
- required = True Specify if free-r flags are must be present (or else generated)

- `test_flag_value = None` This value is usually selected automatically - do not change unless you really know what you're doing!
- `ignore_r_free_flags = False` Use all reflections in refinement (work and test)
- `disable_suitability_test = False`
- `ignore_pdb_hexdigest = False` If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.
- `generate = False` Generate R-free flags (if not available in input files)
- `fraction = 0.1`
- `max_free = 2000`
- `lattice_symmetry_max_delta = 5`
- `use_lattice_symmetry = True`
- `use_dataman_shells = False` Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.
- `n_shells = 20`
- `experimental_phasesfile_name = None` labels = None
- `file_name = None`
- labels = None
- `pdbfile_name = None` Model file(s) name (PDB)
- `file_name = None` Model file(s) name (PDB)
- `monomersfile_name = None` Monomer file(s) name (CIF)
- `file_name = None` Monomer file(s) name (CIF)
- `mapmap_file_name = None` A CCP4-formatted map  
`mapmap_coefficients_file_name = None` MTZ file containing map  
`mapmap_coefficients_label = None` Data label for complex map coefficients in MTZ file
- `map_file_name = None` A CCP4-formatted map
- `d_min = None` Resolution of map
- `map_coefficients_file_name = None` MTZ file containing map
- `map_coefficients_label = None` Data label for complex map coefficients in MTZ file
- `outputfile_name = composite_omit_map.mtz` debug = False Show additional debugging output, and save intermediate files.  
`directory = None` Output directory for GUI jobs  
`job_title = None` Job title in PHENIX GUI, not used on command line
- `file_name = composite_omit_map.mtz`
- debug = False Show additional debugging output, and save intermediate files.
- `directory = None` Output directory for GUI jobs
- `job_title = None` Job title in PHENIX GUI, not used on command line
- `omit_mapmap_type = None` This should be a string representing standard map types, e.g. 2mFo-DFc. You may specify as many of these as you want (each as a separate argument).  
`omit_type = *simple` refine anneal None Determines the method used to prepare the model used for phasing omit maps. The simple option de-biases the phases by treating F(model) directly in an iterative procedure. The refine and anneal options instead run phenix.refine on the partial model(s).  
`fraction_omit = 0.05` Fraction of atoms to omit at

a timeresolution\_factor = 0.25 Grid spacing for assembling partial maps.fill\_missing\_f\_obs = False Fill missing F(obs) with F(calc)exclude\_free\_r\_reflections = True Remove reflections flagged for R-free  
 map\_types = None This should be a space-separated list of strings representing standard map types, e.g. 2mFo-DFc. Identical to the separate map\_type parameter except that map types are combined into a single string.boxingselection = None Selection of atoms to omit in groups. If not specified, the procedure will use the entire structure. Not compatible with omit\_type=simple.single = False Generates a single omit map rather than instead of dividing into boxes and combining individual maps, equivalent to setting fraction\_omit=1.0. The selection must be specified if this option is used.box\_cushion\_radius = 2.5 Additional buffer around omitted region within which to set occupancies to zero.exclude\_bulk\_solvent = False Prevent bulk solvent from extending into omitted regions.optimize\_binning = True Re-group omit regions to reduce the total number of regions while keeping the fraction omitted per region as similar as possible.refinementannealing\_type = \*cartesian torsion annealing\_temperature = Auto Defaults to 5000K for Cartesian, 2500K for torsionncls = Auto Use NCS restraints in refinementharmonic\_restraints = True Apply harmonic restraints to zero-occupancy atoms to prevent other atoms outside the omitted region from filling in the missing scattering.harmonic\_restraints\_sigma = 0.1 vary\_random\_seed = False Vary the random seed between runs.automatic\_linking = False Automatically link covalent ligands (sugars, etc.).debuggingDebugging options, not for general use.flatten\_background = False control\_map = False simplen\_debias\_cycles = 2 neutral\_volume\_box\_cushion\_width = 1 full\_resolution\_map = True

- map\_type = None This should be a string representing standard map types, e.g. 2mFo-DFc. You may specify as many of these as you want (each as a separate argument).
- omit\_type = \*simple refine anneal None Determines the method used to prepare the model used for phasing omit maps. The simple option de-biases the phases by treating F(model) directly in an iterative procedure. The refine and anneal options instead run phenix.refine on the partial model(s).
- fraction\_omit = 0.05 Fraction of atoms to omit at a time
- resolution\_factor = 0.25 Grid spacing for assembling partial maps.
- fill\_missing\_f\_obs = False Fill missing F(obs) with F(calc)
- exclude\_free\_r\_reflections = True Remove reflections flagged for R-free
- map\_types = None This should be a space-separated list of strings representing standard map types, e.g. 2mFo-DFc. Identical to the separate map\_type parameter except that map types are combined into a single string.
- boxingselection = None Selection of atoms to omit in groups. If not specified, the procedure will use the entire structure. Not compatible with omit\_type=simple.single = False Generates a single omit map rather than instead of dividing into boxes and combining individual maps, equivalent to setting fraction\_omit=1.0. The selection must be specified if this option is used.box\_cushion\_radius = 2.5 Additional buffer around omitted region within which to set occupancies to zero.exclude\_bulk\_solvent = False Prevent bulk solvent from extending into omitted regions.optimize\_binning = True Re-group omit regions to reduce the total number of regions while keeping the fraction omitted per region as similar as possible.refinementannealing\_type = \*cartesian torsion annealing\_temperature = Auto Defaults to 5000K for Cartesian, 2500K for torsionncls = Auto Use NCS restraints in refinementharmonic\_restraints = True Apply harmonic restraints to zero-occupancy atoms to prevent other atoms outside the omitted region from filling in the missing scattering.harmonic\_restraints\_sigma = 0.1 vary\_random\_seed = False Vary the random seed between runs.automatic\_linking = False Automatically link covalent ligands (sugars, etc.).debuggingDebugging options, not for general use.flatten\_background = False control\_map = False

- selection = None Selection of atoms to omit in groups. If not specified, the procedure will use the entire structure. Not compatible with omit\_type=simple.
- single = False Generates a single omit map rather than instead of dividing into boxes and combining individual maps, equivalent to setting fraction\_omit=1.0. The selection must be specified if this option is used.
- box\_cushion\_radius = 2.5 Additional buffer around omitted region within which to set occupancies to zero.
- exclude\_bulk\_solvent = False Prevent bulk solvent from extending into omitted regions.
- optimize\_binning = True Re-group omit regions to reduce the total number of regions while keeping the fraction omitted per region as similar as possible.
- refinementannealing\_type = \*cartesian torsion annealing\_temperature = Auto Defaults to 5000K for Cartesian, 2500K for torsion ncs = Auto Use NCS restraints in refinement harmonic\_restraints = True Apply harmonic restraints to zero-occupancy atoms to prevent other atoms outside the omitted region from filling in the missing scattering. harmonic\_restraints\_sigma = 0.1 vary\_random\_seed = False Vary the random seed between runs. automatic\_linking = False Automatically link covalent ligands (sugars, etc.).
- annealing\_type = \*cartesian torsion
- annealing\_temperature = Auto Defaults to 5000K for Cartesian, 2500K for torsion
- ncs = Auto Use NCS restraints in refinement
- harmonic\_restraints = True Apply harmonic restraints to zero-occupancy atoms to prevent other atoms outside the omitted region from filling in the missing scattering.
- harmonic\_restraints\_sigma = 0.1
- vary\_random\_seed = False Vary the random seed between runs.
- automatic\_linking = False Automatically link covalent ligands (sugars, etc.).
- debugging Debugging options, not for general use. flatten\_background = False control\_map = False
- flatten\_background = False
- control\_map = False
- simplen\_debias\_cycles = 2 neutral\_volume\_box\_cushion\_width = 1 full\_resolution\_map = True
- n\_debias\_cycles = 2
- neutral\_volume\_box\_cushion\_width = 1
- full\_resolution\_map = True
- parallelnproc = 1 technology = \*multiprocessing sge lsf pbs condor slurm threading qsub\_command = None
- nproc = 1
- technology = \*multiprocessing sge lsf pbs condor slurm threading
- qsub\_command = None

# Cutting density out of a map with cut\_out\_density

---

cut\_out\_density.html

## Contents

- Author(s)
- Purpose
- Usage
- How cut\_out\_density works:
- Output files from cut\_out\_density
- Examples
- List of all available keywords

## Author(s)

- cut\_out\_density: Tom Terwilliger

## Purpose

This routine cuts density out of a map and writes an MTZ file in space group P1 with map coefficients for the cutout density. The density map is calculated from map coefficients supplied as an input MTZ file. The region to be cut out is specified through the center of the region and dimensions in X Y and Z of a box to be cut out (cutout\_center=X,Y,Z and cutout\_dimensions=a,b,c). Alternatively the region can be specified with a PDB file and optional atom selections, with all points within a specified radius of an atom included, or the region can be specified with a center and the radius of a sphere to include. The density in the region cut out can optionally be offset to have a mean density of zero. The region outside the cut out region always will have a density of zero. The cut out region can be optionally padded with flat density an additional distance beyond the box that is cut out (padding=xx). The resolution of the cutout map can also be specified (the input map will be truncated at the high-resolution limit specified, and the output map will extend to the high-resolution limit.)

## Usage

The main uses of phenix.cut\_out\_density are:

- Cutting out a small part of a map and molecule to work on independently of the rest of the structure. This can be useful if you have a very large structure, as working with a small piece may be much quicker. Note that you can only carry out real-space methods on the extracted map and model.
- Cutting density out of a map to use in molecular replacement. In this case it is easiest to use the keyword "for\_phaser=True" to make a file suitable for use in Phaser.

Please note: you should not use phenix.cut\_out\_density to select regions near your model for display purposes. You can create a highly misleading map by displaying only density that is within a small radius of atoms in your model.

## How cut\_out\_density works:

cut\_out\_density assumes that you are supplying map coefficients and calculates a map. Then it creates a new unit cell in space group P1 with orthogonal axes and dimensions big enough for your cut out region and

padding of zero density around it. The density in your map at the center you specify is then centered at the origin in the new map. Then new map coefficients are created and written out.

You can optionally specify the region to be cut out as a box, a sphere, or a region defined by all points within a specified distance of atoms in a PDB file.

You can specify the center of the region to be cut out with its coordinates, or as the centroid of a PDB file that is input.

If you specify an input PDB file, you can use a selection string to select any part of that model (i.e., one domain or one chain) as the region to be cut out or as the center of the cut out region.

If you specify an input PDB file, then the selected atoms from that file will be offset to match the new map and written out.

As a convenience, you can re-transform a PDB file back to the original position in your original model as well.

## Output files from cut\_out\_density

cutout.mtz: An MTZ file with your cut out density in space group P1.

cutout.pdb: A PDB file with your input molecule translated to match the offset used in creating cutout.mtz

## Examples

Standard run of cut\_out\_density:

Running cut\_out\_density is easy. From the command-line you can type:

```
phenix.cut_out_density model.pdb map_coeffs.mtz
```

Preparing density for molecular replacement in Phaser:

The `for_phaser=True` keyword sets most of the defaults appropriately. You have to provide a PDB file with atoms defining the centers of spheres around which the density should be cut out. Set the `cutout_model_radius` parameter to define the size of the spheres. If you have a reasonably detailed atomic model, a value of 3. may be appropriate, but you may have a PDB file specifying one big sphere as in this command:

```
phenix.cut_out_density for_phaser=True cutout_model_radius=15 sphere.pdb map_coeffs.mtz
```

## List of all available keywords

`input_files``pdb_in` = None Input PDB file marking region to cut out  
`mtz_in` = None Input MTZ file with map coefficients  
`map_coeff_labels` = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: `map_coeff_labels` (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)  
`pdb_to_restore_position` = None You can supply a PDB file that has previously been offset by this routine and it will restore it to its original position. You should specify everything else just as you did originally, then just add these two keywords and their values:

`pdb_to_restore_position=my_offset_pdb_file_with_edits.pdb` and

`pdb_restored_in_position=my_restored_pdb.pdb` Nothing will be run except your `pdb_to_restore_position` will be offset by the right amount and rewritten to `pdb_restored_in_position`

`pdb_to_trim` = None You can supply a PDB file to trim down to the region that is cut out of density. (Note only applicable for box and sphere values of `cutout_type`)  
`output_file``target_output_format` = \*None `pdb` `mmCIF` Desired output format (if possible). Choices

are None ( try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is `mtz_out = cutout.mtz` Output MTZ file in sg p1 with cutout of density `pdb_out = cutout.pdb` Output PDB file offset to match `mtz_out` If `atom_selection` is specified, only selected atoms are written to `cutout.pdb` `pdb_restored_in_position = None` Name of output PDB file restored to its original position. You can supply a PDB file that has previously been offset by this routine and it will restore it to its original position. You should specify everything else just as you did originally, then just add these two keywords and their values:  
`pdb_to_restore_position=my_offset_pdb_file_with_edits.pdb` and  
`pdb_restored_in_position=my_restored_pdb.pdb` Nothing will be run except your `pdb_to_restore_position` will  
`pdb_trimmed = None` Name of output PDB file trimmed to match `cutout.region.log = cutout.log` Output log file  
`params_out = cutout.params.eff` Parameters file to rerun `cutout` `cutout_map_file_name = None` File name for map file if `cutout_as_map=True` `cutout_mask_file_name = None` File name for mask file if  
`cutout_as_map=True` `job_title = None` Job title in PHENIX GUI, not used on command line `directories_output_dir = ""` Output directory where files are to be written  
`gui_output_dir = None` `cutout_region_high_resolution = None` Resolution of output map coeffs (default is `high_resolution_scale` times resolution of input map coeffs.) Purpose is to minimize truncation errors.  
`high_resolution_scale = 0.95` Resolution of output map coeffs will be `high_resolution_scale`\* resolution of input map coeffs if `high_resolution` is None `cutout_center = None` Center of region to be cut out. If `cutout_center=None` and `pdb_in` is specified, the center will be the molecular center of the molecule in `pdb_in` (after applying any atom selections specified with `atom_selection`.) If None and no `pdb_in` is specified, the center will be (0,0,0). `cutout_fix_position = False` If True, then do not translate the density to the origin. Instead leave it in original position. `cutout_dimensions = None` Dimensions (A) of region to be cut out. If None and `pdb_in` is specified, the region will be the the molecular dimensions of the molecule in `pdb_in` (after applying any atom selections specified with `atom_selection`), plus a distance `cutout_model_radius`. If `box_scale` is not defined, the unit cell for the cutout density will have dimensions specified by `cutout_dimensions`, increased by the distance `padding=xxx` on each side. If `cutout_dimensions=10,10,10` and `padding=5`, then the unit cell will be (20,20,20). Otherwise if `box_scale` is defined, the unit cell will be `box_scale` times `cutout_dimensions`. `box_scale = None` If defined, make ...

- `input_files` `pdb_in = None` Input PDB file marking region to cut out `mtz_in = None` Input MTZ file with map coefficients `map_coeff_labels = None` If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: `map_coeff_labels` (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM) `pdb_to_restore_position = None` You can supply a PDB file that has previously been offset by this routine and it will restore it to its original position. You should specify everything else just as you did originally, then just add these two keywords and their values:

`pdb_to_restore_position=my_offset_pdb_file_with_edits.pdb` and

`pdb_restored_in_position=my_restored_pdb.pdb` Nothing will be run except your `pdb_to_restore_position` will be offset by the right amount and rewritten to `pdb_restored_in_position` `pdb_to_trim = None` You can supply a PDB file to trim down to the region that is cut out of density. (Note only applicable for box and sphere values of `cutout_type`)

- `pdb_in = None` Input PDB file marking region to cut out

- `mtz_in = None` Input MTZ file with map coefficients

- `map_coeff_labels = None` If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: `map_coeff_labels` (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)

- `pdb_to_restore_position = None` You can supply a PDB file that has previously been offset by this routine and it will restore it to its original position. You should specify everything else just as you did originally, then just add these two keywords and their values:  
`pdb_to_restore_position=my_offset_pdb_file_with_edits.pdb` and  
`pdb_restored_in_position=my_restored_pdb.pdb` Nothing will be run except your `pdb_to_restore_position` will be offset by the right amount and rewritten to `pdb_restored_in_position`
- `pdb_to_trim = None` You can supply a PDB file to trim down to the region that is cut out of density. (Note only applicable for box and sphere values of `cutout_type`)
- `output_filetarget_output_format = *None` `pdb` `mmCIF` Desired output format (if possible). Choices are `None` ( try to use input format), `pdb`, `mmCIF`. If output model does not fit in `pdb` format, `mmCIF` will be used. Default is `pdb`.  
`mtz_out = cutout.mtz` Output MTZ file in sg p1 with cutout of density  
`pdb_out = cutout.pdb` Output PDB file offset to match `mtz_out` If `atom_selection` is specified, only selected atoms are written to `cutout.pdb`  
`pdb_restored_in_position = None` Name of output PDB file restored to its original position. You can supply a PDB file that has previously been offset by this routine and it will restore it to its original position. You should specify everything else just as you did originally, then just add these two keywords and their values: `pdb_to_restore_position=my_offset_pdb_file_with_edits.pdb` and  
`pdb_restored_in_position=my_restored_pdb.pdb` Nothing will be run except your `pdb_to_restore_position` will  
`pdb_trimmed = None` Name of output PDB file trimmed to match cutout region.  
`log = cutout.log` Output log file  
`params_out = cutout_params.eff` Parameters file to rerun cutout  
`cutout_map_file_name = None` File name for map file if `cutout_as_map=True`  
`cutout_mask_file_name = None` File name for mask file if `cutout_as_map=True`  
`job_title = None` Job title in PHENIX GUI, not used on command line
- `target_output_format = *None` `pdb` `mmCIF` Desired output format (if possible). Choices are `None` ( try to use input format), `pdb`, `mmCIF`. If output model does not fit in `pdb` format, `mmCIF` will be used. Default is `pdb`.
- `mtz_out = cutout.mtz` Output MTZ file in sg p1 with cutout of density
- `pdb_out = cutout.pdb` Output PDB file offset to match `mtz_out` If `atom_selection` is specified, only selected atoms are written to `cutout.pdb`
- `pdb_restored_in_position = None` Name of output PDB file restored to its original position. You can supply a PDB file that has previously been offset by this routine and it will restore it to its original position. You should specify everything else just as you did originally, then just add these two keywords and their values: `pdb_to_restore_position=my_offset_pdb_file_with_edits.pdb` and  
`pdb_restored_in_position=my_restored_pdb.pdb` Nothing will be run except your `pdb_to_restore_position` will
- `pdb_trimmed = None` Name of output PDB file trimmed to match cutout region.
- `log = cutout.log` Output log file
- `params_out = cutout_params.eff` Parameters file to rerun cutout
- `cutout_map_file_name = None` File name for map file if `cutout_as_map=True`
- `cutout_mask_file_name = None` File name for mask file if `cutout_as_map=True`
- `job_title = None` Job title in PHENIX GUI, not used on command line
- `directoriesoutput_dir = ""` Output directory where files are to be written  
`gui_output_dir = None`
- `output_dir = ""` Output directory where files are to be written
- `gui_output_dir = None`

- `cutout_region` `high_resolution = None` Resolution of output map coeffs (default is `high_resolution_scale` times resolution of input map coeffs.) Purpose is to minimize truncation errors. `high_resolution_scale = 0.95` Resolution of output map coeffs will be `high_resolution_scale * resolution of input map coeffs` if `high_resolution` is `None`. `cutout_center = None` Center of region to be cut out. If `cutout_center = None` and `pdb_in` is specified, the center will be the molecular center of the molecule in `pdb_in` (after applying any atom selections specified with `atom_selection`.) If `None` and no `pdb_in` is specified, the center will be (0,0,0). `cutout_fix_position = False` If `True`, then do not translate the density to the origin. Instead leave it in original position. `cutout_dimensions = None` Dimensions (Å) of region to be cut out. If `None` and `pdb_in` is specified, the region will be the molecular dimensions of the molecule in `pdb_in` (after applying any atom selections specified with `atom_selection`), plus a distance `cutout_model_radius`. If `box_scale` is not defined, the unit cell for the cutout density will have dimensions specified by `cutout_dimensions`, increased by the distance `padding = xxx` on each side. If `cutout_dimensions = 10,10,10` and `padding = 5`, then the unit cell will be (20,20,20). Otherwise if `box_scale` is defined, the unit cell will be `box_scale` times `cutout_dimensions`. `box_scale = None` If defined, make the dimensions of the final cell `box_scale` times `cutout_dimensions`. `padding = 5` Padding of flat density to add around cutout density. The unit cell for the cutout density will have dimensions specified by `cutout_dimensions`, increased by the distance `padding = xxx` on each side. If `cutout_dimensions` is `None`, then the unit cell will be the molecular dimensions of the molecule in `pdb_in`, increased by `padding` on each side. If `cutout_dimensions = 10,10,10` and `padding = 5`, then the unit cell will be approximately (20,20,20). Note that the unit cell dimensions may differ from this slightly as each must fall on an exact grid multiple of 12 and the cutout region will determine the grid. `cutout_type = *box` sphere model If `box` and `cutout_dimensions` are specified, actual density will be replaced with flat density added outside the box defined with `cutout_dimensions`. If `box` and `pdb_in` are specified then the cut out region will be a box defined by the model plus a distance `cutout_model_radius`. If `sphere` is specified, then flat density will replace actual density outside of a sphere with radius `cutout_sphere_radius`. `cutout_dimensions` will be the dimensions of the sphere. If `model` and `pdb_in` are specified then the cut out region will be the region defined by the model plus a distance `cutout_model_radius`. Flat density will replace actual density outside of region within `cutout_model_radius` of an atom in `pdb_in`. `cutout_sphere_radius = 10`. If `cutout_type` is `sphere`, then flat density will replace actual density outside of a sphere with radius `cutout_sphere_radius`. `soft_mask = False` If `True` and `cutout_type` is `model`, then create soft mask around density (Gaussian smoothing between density inside and outside mask). This will also increase the size of the box, sphere or region by about 2 times the `buffer_radius`, or by default 2 times the resolution of the map. Requires `high_resolution` to be set. `buffer_radius = None` Buffer radius for mask smoothing. Default is resolution. `map_scale = 1`. You can scale the output map if you want to. `cutout_model_radius = 5`. If `cutout_type` is `model`, then flat density will replace actual density outside of region within `cutout_model_radius` of an atom in `pdb_in` (after application of any selections specified with `atom_selection`). `cutout_subtract_mean = False` If `True`, then mean density in cutout region will be calculated and subtracted from all the density in that region, yielding a mean of 0. This is recommended if the density is to be used in molecular replacement. `include_hetatm = True` If `True`, then include HETATM atoms in input PDB file in defining density...

- `high_resolution = None` Resolution of output map coeffs (default is `high_resolution_scale` times resolution of input map coeffs.) Purpose is to minimize truncation errors.

- `high_resolution_scale = 0.95` Resolution of output map coeffs will be `high_resolution_scale * resolution of input map coeffs` if `high_resolution` is `None`

- `cutout_center = None` Center of region to be cut out. If `cutout_center = None` and `pdb_in` is specified, the center will be the molecular center of the molecule in `pdb_in` (after applying any atom selections specified with `atom_selection`.) If `None` and no `pdb_in` is specified, the center will be (0,0,0).

- `cutout_fix_position` = False If True, then do not translate the density to the origin. Instead leave it in original position.
- `cutout_dimensions` = None Dimensions (A) of region to be cut out. If None and `pdb_in` is specified, the region will be the molecular dimensions of the molecule in `pdb_in` (after applying any atom selections specified with `atom_selection`), plus a distance `cutout_model_radius`. If `box_scale` is not defined, the unit cell for the cutout density will have dimensions specified by `cutout_dimensions`, increased by the distance `padding=xxx` on each side. If `cutout_dimensions=10,10,10` and `padding=5`, then the unit cell will be (20,20,20). Otherwise if `box_scale` is defined, the unit cell will be `box_scale` times `cutout_dimensions`.
- `box_scale` = None If defined, make the dimensions of the final cell `box_scale` times `cutout_dimensions`.
- `padding` = 5 Padding of flat density to add around cutout density. The unit cell for the cutout density will have dimensions specified by `cutout_dimensions`, increased by the distance `padding=xxx` on each side. If `cutout_dimensions` is None, then the unit cell will be the molecular dimensions of the molecule in `pdb_in`, increased by `padding` on each side. If `cutout_dimensions=10,10,10` and `padding=5`, then the unit cell will be approximately (20,20,20). Note that the unit cell dimensions may differ from this slightly as each must fall on an exact grid multiple of 12 and the cutout region will determine the grid.
- `cutout_type` = \*box sphere model If box and `cutout_dimensions` are specified, actual density will be replaced with flat density added outside the box defined with `cutout_dimensions`. If box and `pdb_in` are specified then the cut out region will be a box defined by the model plus a distance `cutout_model_radius`. If sphere is specified, then flat density will replace actual density outside of a sphere with radius `cutout_sphere_radius`. `cutout_dimensions` will be the dimensions of the sphere. If model and `pdb_in` are specified then the cut out region will be the region defined by the model plus a distance `cutout_model_radius`. Flat density will replace actual density outside of region within `cutout_model_radius` of an atom in `pdb_in`.
- `cutout_sphere_radius` = 10. If `cutout_type` is sphere, then flat density will replace actual density outside of a sphere with radius `cutout_sphere_radius`.
- `soft_mask` = False If True and `cutout_type` is model, then create soft mask around density (Gaussian smoothing between density inside and outside mask). This will also increase the size of the box, sphere or region by about 2 times the `buffer_radius`, or by default 2 times the resolution of the map. Requires `high_resolution` to be set.
- `buffer_radius` = None Buffer radius for mask smoothing. Default is resolution
- `map_scale` = 1. You can scale the output map if you want to.
- `cutout_model_radius` = 5. If `cutout_type` is model, then flat density will replace actual density outside of region within `cutout_model_radius` of an atom in `pdb_in` (after application of any selections specified with `atom_selection`).
- `cutout_subtract_mean` = False If True, then mean density in cutout region will be calculated and subtracted from all the density in that region, yielding a mean of 0. This is recommended if the density is to be used in molecular replacement.
- `include_hetatm` = True If True, then include HETATM atoms in input PDB file in defining density to be included (e.g., if `cutout_type` is model)
- `atom_selection` = None Any selection specified with `atom_selection` is applied to input model before using the model.
- `cutout_as_map` = False Return cut out region as map (alternative is map coeffs).
- `controlverbose` = False Verbose output
- `raise_sorry` = False Raise sorry if problems
- `debug` = False Set debugging output
- `dry_run` = False Just read in and check parameter names
- `for_phaser` = False Set

defaults for phaser: cutout\_fix\_position=True, cutout\_type=model, box\_scale=3.0 (if not set explicitly), high\_resolution is set to allow molecular transform to be interpolated to resolution of original map  
mapresolve\_command\_list = None You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (&quot;) and each individual command in single quotes (') like this: resolve\_command\_list=&quot;'no\_build' 'b\_overall 23' &quot;;

- verbose = False Verbose output
- raise\_sorry = False Raise sorry if problems
- debug = False Debugging output
- dry\_run = False Just read in and check parameter names
- for\_phaser = False Set defaults for phaser: cutout\_fix\_position=True, cutout\_type=model, box\_scale=3.0 (if not set explicitly), high\_resolution is set to allow molecular transform to be interpolated to resolution of original map
- resolve\_command\_list = None You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (&quot;) and each individual command in single quotes (') like this: resolve\_command\_list=&quot;'no\_build' 'b\_overall 23' &quot;;

# Merging two models with density\_outside\_model

---

density\_outside\_model.html

## Author(s)

- density\_outside\_model: Tom Terwilliger

## Purpose

This routine takes a model and map coefficients and returns new map coefficients with density set to zero near the atoms in the model

## Usage

The main uses of phenix.density\_outside\_model are:

- Creating a map where all the parts of the map that have been explained by a model are removed.

## How density\_outside\_model works:

Density\_outside\_model calculates a map from the map coefficients. Then all map values at points near an atom in the model are set to zero. Then a new set of map coefficients is calculated and written out.

Optionally the model can be trimmed so that the last n\_trim residues in each segment of the model are not included.

## Output files from density\_outside\_model

density\_removed.mtz: An MTZ file with your new map coefficients.

## Examples

### Standard run of density\_outside\_model:

Running density\_outside\_model is easy. From the command-line you can type:

```
phenix.density_outside_model pdb_in=mask_pdb.pdb \  
map_coeffs.mtz
```

## Possible Problems

### Specific limitations and problems:

## Literature

## Additional information

## List of all available keywords

density\_outside\_modelinput\_files`pdb_in = None` Input PDB file marking region to remove from map  
`mtz_in = None` Input MTZ file with map coefficients  
`map_coeff_labels = None` If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: `map_coeff_labels` (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)  
`output_files`  
`mtz_out = density_removed.mtz` Output MTZ file in sg p1 with density\_removed of density  
`log = density_removed.log` Output log file  
`params_out = density_removed_params.eff` Parameters file to rerun density\_removed  
`directory`  
`temp_dir = "temp_dir"` Optional temporary work directory  
`output_dir = ""` Output directory where files are to be written  
`density_removed_region`  
`density_removed_model_radius = 2.5` If density\_removed\_type is model, then flat density will replace actual density inside of region within density\_removed\_model\_radius of an atom in `pdb_in` (after application of any selections specified with `atom_selection`).  
`include_hetatm = True` If True, then include HETATM atoms in input PDB file in defining region to be included  
`atom_selection = None` Any selection specified with `atom_selection` is applied to input model before using the model.  
`n_trim = 0` Trim all ends of segments by `n_trim` residues before calculating density  
`density_removed_as_map = False` Return modified density as map (alternative is map coeffs).  
`chain_type = PROTEIN` chain type (PROTEIN RNA DNA)  
`control`  
`verbose = False` Verbose output  
`raise_sorry = False` Raise sorry if problems  
`debug = False` Debugging output  
`dry_run = False` Just read in and check parameter names  
`resolve_command_list = None` You can supply any resolve command here  
 NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (&quot;) and each individual command in single quotes (') like this:  
`resolve_command_list=&quot;'no_build' 'b_overall 23' &quot;`;

- density\_outside\_modelinput\_files`pdb_in = None` Input PDB file marking region to remove from map  
`mtz_in = None` Input MTZ file with map coefficients  
`map_coeff_labels = None` If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: `map_coeff_labels` (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)  
`output_files`  
`mtz_out = density_removed.mtz` Output MTZ file in sg p1 with density\_removed of density  
`log = density_removed.log` Output log file  
`params_out = density_removed_params.eff` Parameters file to rerun density\_removed  
`directory`  
`temp_dir = "temp_dir"` Optional temporary work directory  
`output_dir = ""` Output directory where files are to be written  
`density_removed_region`  
`density_removed_model_radius = 2.5` If density\_removed\_type is model, then flat density will replace actual density inside of region within density\_removed\_model\_radius of an atom in `pdb_in` (after application of any selections specified with `atom_selection`).  
`include_hetatm = True` If True, then include HETATM atoms in input PDB file in defining region to be included  
`atom_selection = None` Any selection specified with `atom_selection` is applied to input model before using the model.  
`n_trim = 0` Trim all ends of segments by `n_trim` residues before calculating density  
`density_removed_as_map = False` Return modified density as map (alternative is map coeffs).  
`chain_type = PROTEIN` chain type (PROTEIN RNA DNA)  
`control`  
`verbose = False` Verbose output  
`raise_sorry = False` Raise sorry if problems  
`debug = False` Debugging output  
`dry_run = False` Just read in and check parameter names  
`resolve_command_list = None` You can supply any resolve command here  
 NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (&quot;) and each individual command in single quotes (') like this:  
`resolve_command_list=&quot;'no_build' 'b_overall 23' &quot;`;

- input\_files`pdb_in = None` Input PDB file marking region to remove from map  
`mtz_in = None` Input MTZ file with map coefficients  
`map_coeff_labels = None` If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify:

map\_coeff\_labels (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)

- pdb\_in = None Input PDB file marking region to remove from map
- mtz\_in = None Input MTZ file with map coefficients
- map\_coeff\_labels = None If map coefficients cannot be identified automatically from your MTZ file, you can specify the label or labels for them. (Please separate labels with blank space, MTZ columns grouped together separated by commas with no blanks.) You can specify: map\_coeff\_labels (e.g., FWT,PHIFWT) amplitudes and phases (e.g., FP,SIGFP PHIB) or amplitudes, phases, weights (e.g., FP,SIGFP PHIB FOM)
- output\_filesmtz\_out = density\_removed.mtz Output MTZ file in sg p1 with density\_removed of densitylog = density\_removed.log Output log fileparams\_out = density\_removed\_params.eff Parameters file to rerun density\_removed
- mtz\_out = density\_removed.mtz Output MTZ file in sg p1 with density\_removed of density
- log = density\_removed.log Output log file
- params\_out = density\_removed\_params.eff Parameters file to rerun density\_removed
- directoriestemp\_dir = "temp\_dir" Optional temporary work directory output\_dir = "" Output directory where files are to be written
- temp\_dir = "temp\_dir" Optional temporary work directory
- output\_dir = "" Output directory where files are to be written
- density\_removed\_regiondensity\_removed\_model\_radius = 2.5 If density\_removed\_type is model, then flat density will replace actual density inside of region within density\_removed\_model\_radius of an atom in pdb\_in (after application of any selections specified with atom\_selection.include\_hetatm = True If True, then include HETATM atoms in input PDB file in defining region to be included atom\_selection = None Any selection specified with atom\_selection is applied to input model before using the model.n\_trim = 0 Trim all ends of segments by n\_trim residues before calculating densitydensity\_removed\_as\_map = False Return modified density as map (alternative is map coeffs).chain\_type = PROTEIN chain type (PROTEIN RNA DNA)
- density\_removed\_model\_radius = 2.5 If density\_removed\_type is model, then flat density will replace actual density inside of region within density\_removed\_model\_radius of an atom in pdb\_in (after application of any selections specified with atom\_selection.
- include\_hetatm = True If True, then include HETATM atoms in input PDB file in defining region to be included
- atom\_selection = None Any selection specified with atom\_selection is applied to input model before using the model.
- n\_trim = 0 Trim all ends of segments by n\_trim residues before calculating density
- density\_removed\_as\_map = False Return modified density as map (alternative is map coeffs).
- chain\_type = PROTEIN chain type (PROTEIN RNA DNA)
- controlverbose = False Verbose outputraise\_sorry = False Raise sorry if problemsdebug = False Debugging outputdry\_run = False Just read in and check parameter namesresolve\_command\_list = None You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (&quot;) and each individual command in single quotes (') like this: resolve\_command\_list=&quot;'no\_build' 'b\_overall 23' &quot;;
- verbose = False Verbose output

- `raise_sorry = False` Raise sorry if problems
- `debug = False` Debugging output
- `dry_run = False` Just read in and check parameter names
- `resolve_command_list = None` You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (&quot;) and each individual command in single quotes (') like this: `resolve_command_list=&quot;'no_build' 'b_overall 23' &quot;`

# phenix.fem: compute Feature Enhanced Map (FEM)

---

fem.html

## Description

The program reads PDB model, reflection data (Fobs or lobs), experimental uncertainties and R-free flags, if available. Then it computes {2mFobs-DFmodel, Phase\_model} map that is subject to FEM procedure. Output is MTZ file with three sets of Fourier map coefficients: 2mFoDFc,PHI2mFoDFc: usual {2mFobs-DFmodel, Phase\_model} sigma-a weighted synthesis. No anisotropy correction or filling missing data. ResolveDM,PHIResolveDM: Resolve density modified map coefficients FEM,PHIFEM: Feature Enhanced Map

The program reads PDB model, reflection data (Fobs or lobs), experimental uncertainties and R-free flags, if available. Then it computes {2mFobs-DFmodel, Phase\_model} map that is subject to FEM procedure.

Output is MTZ file with three sets of Fourier map coefficients:

2mFoDFc,PHI2mFoDFc: usual {2mFobs-DFmodel, Phase\_model} sigma-a weighted synthesis. No anisotropy correction or filling missing data. ResolveDM,PHIResolveDM: Resolve density modified map coefficients FEM,PHIFEM: Feature Enhanced Map

- 2mFoDFc,PHI2mFoDFc: usual {2mFobs-DFmodel, Phase\_model} sigma-a weighted synthesis. No anisotropy correction or filling missing data.
- ResolveDM,PHIResolveDM: Resolve density modified map coefficients
- FEM,PHIFEM: Feature Enhanced Map

## Command line usage examples

Running with all defaults: % phenix.fem model.pdb data.mtz Specify data label: % phenix.fem data.mtz model.pdb label=Fobs Specify scattering dictionary (X-ray, electron, neutron): % phenix.fem model.pdb data.mtz scattering\_label=neutron Some other settings and their defaults: use\_omit=True : use OMIT map use\_resolve=Auto : use Resolve DM as extra filter use\_max\_map=True : use 'maximal synthesis' file\_name\_prefix=fem : file prefix for output For more settings run phenix.fem without arguments in the command line.

- Running with all defaults: % phenix.fem model.pdb data.mtz

Running with all defaults:

```
% phenix.fem model.pdb data.mtz
```

- Specify data label: % phenix.fem data.mtz model.pdb label=Fobs

Specify data label:

```
% phenix.fem data.mtz model.pdb label=Fobs
```

- Specify scattering dictionary (X-ray, electron, neutron): % phenix.fem model.pdb data.mtz scattering\_label=neutron

Specify scattering dictionary (X-ray, electron, neutron):

```
% phenix.fem model.pdb data.mtz scattering_label=neutron
```

- Some other settings and their defaults: use\_omit=True : use OMIT map use\_resolve=Auto : use Resolve DM as extra filter use\_max\_map=True : use 'maximal synthesis' file\_name\_prefix=fem : file prefix for output For more settings run phenix.fem without arguments in the command line.

Some other settings and their defaults:

use\_omit=True : use OMIT map use\_resolve=Auto : use Resolve DM as extra filter use\_max\_map=True : use 'maximal synthesis' file\_name\_prefix=fem : file prefix for output For more settings run phenix.fem without arguments in the command line.

- use\_omit=True : use OMIT map
- use\_resolve=Auto : use Resolve DM as extra filter
- use\_max\_map=True : use 'maximal synthesis'
- file\_name\_prefix=fem : file prefix for output
- For more settings run phenix.fem without arguments in the command line.

## GUI

Graphical user interface is available

Additional information

Latest presentation of FEM methodology: [http://phenix-online.org/presentations/fem\\_06MAY2014.pdf](http://phenix-online.org/presentations/fem_06MAY2014.pdf)

Publication in Computational Crystallography Newsletter (front page):

[http://www.phenix-online.org/newsletter/CCN\\_2013\\_07.pdf](http://www.phenix-online.org/newsletter/CCN_2013_07.pdf)

Latest presentation of FEM methodology:

[http://phenix-online.org/presentations/fem\\_06MAY2014.pdf](http://phenix-online.org/presentations/fem_06MAY2014.pdf)

Publication in Computational Crystallography Newsletter (front page):

[http://www.phenix-online.org/newsletter/CCN\\_2013\\_07.pdf](http://www.phenix-online.org/newsletter/CCN_2013_07.pdf)

## References

FEM: feature-enhanced map. P.V. Afonine, N.W. Moriarty, M. Mustyakimov, O.V. Sobolev, T.C. Terwilliger, D. Turk, A. Urzhumtsev, and P.D. Adams. Acta Cryst. D71, 646-666 (2015).

- FEM: feature-enhanced map. P.V. Afonine, N.W. Moriarty, M. Mustyakimov, O.V. Sobolev, T.C. Terwilliger, D. Turk, A. Urzhumtsev, and P.D. Adams. Acta Cryst. D71, 646-666 (2015).

## List of all available keywords

random\_seed = 2679941 use\_omit = True Use composite OMIT protocolsharp = True use\_unsharp\_masking = True resolution\_factor = 1./4 signal\_threshold = 0.5 use\_resolve = Auto use\_max\_map = True ignore\_zero\_occupancy\_atoms = True job\_title = None Job title in PHENIX GUI, not used on command lineinputsequence = None scattering\_table = wk1995 it1992 \*n\_gaussian neutron electron wavelength = None energy = None use\_experimental\_phases = False xray\_dataScope of X-ray data and free-R flagsfile\_name = None labels = None high\_resolution = None low\_resolution = None twin\_law = None outliers\_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliersfrench\_wilson\_scale = True sigma\_fobs\_rejection\_criterion = None sigma\_iobs\_rejection\_criterion = None ignore\_all\_zeros = True force\_anomalous\_flag\_to\_be\_equal\_to = None convert\_to\_non\_anomalous\_if\_ratio\_pairs\_lone\_less\_than\_threshold = 0.5 french\_wilsonmax\_bins = 60 Maximum number of resolution binsmin\_bin\_size = 40 Minimum number of reflections per binr\_free\_flagsfile\_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None required = True Specify if free-r flags are must be present (or else generated)test\_flag\_value = None This value is usually selected automatically - do not change unless you

really know what you're doing!ignore\_r\_free\_flags = False Use all reflections in refinement (work and test)disable\_suitability\_test = False ignore\_pdb\_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max\_free = 2000 lattice\_symmetry\_max\_delta = 5 use\_lattice\_symmetry = True use\_dataman\_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n\_shells = 20 pdbfile\_name = None Model file(s) name (PDB)monomersfile\_name = None Monomer file(s) name (CIF)mapsmap\_file\_name = None A CCP4-formatted mapd\_min = None Resolution of mapmap\_coefficients\_file\_name = None MTZ file containing mapmap\_coefficients\_label = None Data label for complex map coefficients in MTZ fileoutputfile\_name\_prefix = fem column\_root\_label = FEM gui\_output\_dir = None

- random\_seed = 2679941
- use\_omit = True Use composite OMIT protocol
- sharp = True
- use\_unsharp\_masking = True
- resolution\_factor = 1./4
- signal\_threshold = 0.5
- use\_resolve = Auto
- use\_max\_map = True
- ignore\_zero\_occupancy\_atoms = True
- job\_title = None Job title in PHENIX GUI, not used on command line
- inputsequence = None scattering\_table = wk1995 it1992 \*n\_gaussian neutron electron wavelength = None energy = None use\_experimental\_phases = False xray\_dataScope of X-ray data and free-R flagsfile\_name = None labels = None high\_resolution = None low\_resolution = None twin\_law = None outliers\_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliersfrench\_wilson\_scale = True sigma\_fobs\_rejection\_criterion = None sigma\_iobs\_rejection\_criterion = None ignore\_all\_zeros = True force\_anomalous\_flag\_to\_be\_equal\_to = None convert\_to\_non\_anomalous\_if\_ratio\_pairs\_lone\_less\_than\_threshold = 0.5 french\_wilsonmax\_bins = 60 Maximum number of resolution binsmin\_bin\_size = 40 Minimum number of reflections per binr\_free\_flagsfile\_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None required = True Specify if free-r flags are must be present (or else generated)test\_flag\_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore\_r\_free\_flags = False Use all reflections in refinement (work and test)disable\_suitability\_test = False ignore\_pdb\_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max\_free = 2000 lattice\_symmetry\_max\_delta = 5 use\_lattice\_symmetry = True use\_dataman\_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n\_shells = 20 pdbfile\_name = None Model file(s) name (PDB)monomersfile\_name = None Monomer file(s) name (CIF)mapsmap\_file\_name = None A CCP4-formatted mapd\_min = None Resolution of mapmap\_coefficients\_file\_name = None MTZ file containing mapmap\_coefficients\_label = None Data label for complex map coefficients in MTZ file
- sequence = None
- scattering\_table = wk1995 it1992 \*n\_gaussian neutron electron
- wavelength = None
- energy = None

- use\_experimental\_phases = False
- xray\_dataScope of X-ray data and free-R flagsfile\_name = None labels = None high\_resolution = None low\_resolution = None twin\_law = None outliers\_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliersfrench\_wilson\_scale = True sigma\_fobs\_rejection\_criterion = None sigma\_iobs\_rejection\_criterion = None ignore\_all\_zeros = True force\_anomalous\_flag\_to\_be\_equal\_to = None convert\_to\_non\_anomalous\_if\_ratio\_pairs\_lone\_less\_than\_threshold = 0.5 french\_wilsonmax\_bins = 60 Maximum number of resolution binsmin\_bin\_size = 40 Minimum number of reflections per binr\_free\_flagsfile\_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None required = True Specify if free-r flags are must be present (or else generated)test\_flag\_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore\_r\_free\_flags = False Use all reflections in refinement (work and test)disable\_suitability\_test = False ignore\_pdb\_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max\_free = 2000 lattice\_symmetry\_max\_delta = 5 use\_lattice\_symmetry = True use\_dataman\_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n\_shells = 20
- file\_name = None
- labels = None
- high\_resolution = None
- low\_resolution = None
- twin\_law = None
- outliers\_rejection = True Remove basic wilson outliers , extreme wilson outliers , and beamstop shadow outliers
- french\_wilson\_scale = True
- sigma\_fobs\_rejection\_criterion = None
- sigma\_iobs\_rejection\_criterion = None
- ignore\_all\_zeros = True
- force\_anomalous\_flag\_to\_be\_equal\_to = None
- convert\_to\_non\_anomalous\_if\_ratio\_pairs\_lone\_less\_than\_threshold = 0.5
- french\_wilsonmax\_bins = 60 Maximum number of resolution binsmin\_bin\_size = 40 Minimum number of reflections per bin
- max\_bins = 60 Maximum number of resolution bins
- min\_bin\_size = 40 Minimum number of reflections per bin
- r\_free\_flagsfile\_name = None This is normally the same as the file containing Fobs and is usually selected automatically.label = None required = True Specify if free-r flags are must be present (or else generated)test\_flag\_value = None This value is usually selected automatically - do not change unless you really know what you're doing!ignore\_r\_free\_flags = False Use all reflections in refinement (work and test)disable\_suitability\_test = False ignore\_pdb\_hexdigest = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.generate = False Generate R-free flags (if not available in input files)fraction = 0.1 max\_free = 2000 lattice\_symmetry\_max\_delta = 5 use\_lattice\_symmetry = True use\_dataman\_shells = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.n\_shells = 20

- `file_name` = None This is normally the same as the file containing Fobs and is usually selected automatically.
- `label` = None
- `required` = True Specify if free-r flags are must be present (or else generated)
- `test_flag_value` = None This value is usually selected automatically - do not change unless you really know what you're doing!
- `ignore_r_free_flags` = False Use all reflections in refinement (work and test)
- `disable_suitability_test` = False
- `ignore_pdb_hexdigest` = False If True, disables safety check based on MD5 hexdigests stored in PDB files produced by previous runs.
- `generate` = False Generate R-free flags (if not available in input files)
- `fraction` = 0.1
- `max_free` = 2000
- `lattice_symmetry_max_delta` = 5
- `use_lattice_symmetry` = True
- `use_dataman_shells` = False Used to avoid biasing of the test set by certain types of non-crystallographic symmetry.
- `n_shells` = 20
- `pdbfile_name` = None Model file(s) name (PDB)
- `file_name` = None Model file(s) name (PDB)
- `monomersfile_name` = None Monomer file(s) name (CIF)
- `file_name` = None Monomer file(s) name (CIF)
- `mapmap_file_name` = None A CCP4-formatted map
- `mapmap_min` = None Resolution of map
- `mapmap_coefficients_file_name` = None MTZ file containing map
- `mapmap_coefficients_label` = None Data label for complex map coefficients in MTZ file
- `map_file_name` = None A CCP4-formatted map
- `d_min` = None Resolution of map
- `map_coefficients_file_name` = None MTZ file containing map
- `map_coefficients_label` = None Data label for complex map coefficients in MTZ file
- `outputfile_name_prefix` = fem `column_root_label` = FEM `gui_output_dir` = None
- `file_name_prefix` = fem
- `column_root_label` = FEM
- `gui_output_dir` = None

# Finding NCS from density in a map with `find_ncs_from_density`

---

`find_ncs_from_density.html`

## Author(s)

- `find_ncs_from_density`: Tom Terwilliger

## Purpose

`find_ncs_from_density` is a tool to identify NCS relationships directly from an electron density map specified with a map coefficients file. It writes out the NCS operators in a format that can be read by `find_ncs` and by the PHENIX wizards. Usually `find_ncs_from_density` is used through `find_ncs` (`phenix.find_ncs` contains `find_ncs_from_density`).

## Usage

### How `find_ncs_from_density` works:

- The `find_ncs_from_density` algorithm first identifies potential locations of centers of macromolecules in the density map by finding maxima of the local RMS density using `phenix.guess_molecular_centers`.
- Then it cuts out a sphere of density centered at a trial center and carries out an FFT-based rotation-translation search to find all occurrences of similar density in the asymmetric unit of your map.
- The region over which NCS-related correlation is high is identified and the operators are written out as a "`find_ncs.ncs_spec`" file that can be read by the phenix wizards and a "`find_ncs.phenix_refine`" file that can be read by `phenix.refine`.
- You can run the algorithm as a whole or you can run each part separately with `phenix.guess_molecular_centers` and `phenix.find_ncs_from_density`.

## Output files from `find_ncs_from_density`

`find_ncs_from_density.ncs_spec`: A formatted file that can be read by `phenix.find_ncs` and the PHENIX wizards that specifies the NCS operators. `temp_dir/ncs_avg_map_coeffs.mtz`: This is an mtz file with just one averaged NCS region written out in space group P1.

## Examples

### Standard run of `find_ncs_from_density`:

Running `find_ncs_from_density` is easy. From the command-line you can type:

```
phenix.find_ncs_from_density map_coeffs.mtz
```

## Possible Problems

If your molecule is large, you may wish to increase the radius used for NCS identification (`density_radius`) and the peak separation (`peak_separation`). If your molecule has pseudo-symmetry you may also want to increase these parameters.

If you know how many NCS copies are present, you may wish to specifying that with `ncs_copies_max`.

If you want to have the program look much harder for NCS, try the keyword `n_center_use=20` (or some other number larger than 1). It will then take the top 20 positions found with `guess_molecular_centers` and try each one as a center for a molecule (instead of trying just the top 1 position).

## Literature

Finding non-crystallographic symmetry in density maps of macromolecular structures. T.C. Terwilliger. J Struct Funct Genomics 14, 91-5 (2013).

- Finding non-crystallographic symmetry in density maps of macromolecular structures. T.C. Terwilliger. J Struct Funct Genomics 14, 91-5 (2013).

## List of all available keywords

`find_ncs_from_densityinput_filesmtz_in` = None MTZ file with coefficients for a map `labin` = "" Labin line for MTZ file with map coefficients. This is optional if `find_ncs_from_density` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where `myFP` is your column label for FP `center_pdb_in` = None Optional PDB file with list of centers of density to be used as search model `density_mtz_in` = None MTZ file with coefficients for density around center defined by first atom in `center_pdb_in` (required) NOTE: Must be in space group p1 with identical cell to `mtz_in` `density_labin` = "" Labin line for MTZ file `density_mtz_in`. This is optional if `find_ncs_from_density` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where `myFP` is your column label for FP `output_filescenters_pdb_out` = `guess_molecular_centers.pdb` Output PDB file with coords of centers `log` = `find_ncs_from_density.log` Output log file `ncs_spec_file` = 'find\_ncs\_from\_density.ncs\_spec' NCS specification file with all NCS information `params_out` = `find_ncs_from_density_params.eff` Parameters file to rerun `find_ncs_from_density` `ncs_mask_file` = None If defined, map coefficients will be written to this file that show the asymmetric unit of NCS. `directory` `temp_dir` = "temp\_dir" Temporary work directory `output_dir` = "" Output directory where files are to be written `density_search` `density_radius` = 10. Radius for density to be cut out and compared `peak_separation` = 15 Minimum distance between centers. Use about  $1.5 * \text{density\_radius}$  `density_peaks` = 20 Number of NCS-related peaks of density to output `delta_phi` = 20 Angular spacing of search `rotz_only` = False Rotate around Z only `ncs_copies_max` = None Maximum number of NCS copies to look for `min_ratio_to_top_cc` = 0.75 Peaks will be kept up to `ncs_copies_max`, or `min_ratio_to_top_cc` \* best cc, whichever comes first `minimum_ncs_cc_from_density` = 0.30 Overall NCS CC at full resolution must be at least this high `fraction_ncs_min_from_density` = 0.05 Minimum fraction of the asymmetric unit covered by NCS. NCS ignored if less than this. `dump_ncs_density` = False You can dump 1 mtz file for each possible NCS copy so that you can see if the NCS is real Each file has the local map transformed to the orientation and position of copy 1 so they should all superimpose The files are written to the `temp_dir` (e.g., `temp_dir/ncs_1_map_coeffs_001.mtz`) `map_operators_inside_unit_cell` = False If set then put the operators inside the unit cell. This removes the use of space-group symmetry for SG P1 and is used normally for non-crystal data. `find_centers` `smoothing_radius` = None Radius for smoothing squared density to find centers Choose a smaller value to get more center guesses. Default is same value as `density_radius`. `n_center_find` = None Target number of centers to find Smoothing radius will be varied from 2 to  $2 * \text{smoothing\_radius}$  and the value giving the number of peaks closest to `n_center_find` will be used. Normally leave this set to None `n_center_use` = 20 Number of locations to consider as molecular centers in density search. Try increasing this number to look harder for NCS. `try_all_centers` = False If set, try all `n_center_use` centers. Normally stop if a good solution is found. `ok_ncs_cc` = 0.6 NCS correlation good enough to terminate search `crystal_info` `resolution` = 4. High-resolution limit for map calculation It is useful to cut the resolution at

3-5 Å If you use higher resolution, use a finer `delta_phi` solvent fraction `control` verbose = True Verbose output `debug` = False Debugging output `raise_sorry` = False Raise sorry if problems `dry_run` = False Just read in and check parameter names `resolve_command_list` = None You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (&quot;,) and each individual command in single quotes (') like this:  
`resolve_command_list=&quot;'no_build...`

- `find_ncs_from_density` input\_files `mtz_in` = None MTZ file with coefficients for a map `labin` = "" Labin line for MTZ file with map coefficients. This is optional if `find_ncs_from_density` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where `myFP` is your column label for FP `center_pdb_in` = None Optional PDB file with list of centers of density to be used as search model `density_mtz_in` = None MTZ file with coefficients for density around center defined by first atom in `center_pdb_in` (required) NOTE: Must be in space group `p1` with identical cell to `mtz_in` `density_labin` = "" Labin line for MTZ file `density_mtz_in`. This is optional if `find_ncs_from_density` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where `myFP` is your column label for FP  
`output_files` `centers_pdb_out` = `guess_molecular_centers.pdb` Output PDB file with coords of centers `log` = `find_ncs_from_density.log` Output log file `ncs_spec_file` = 'find\_ncs\_from\_density.ncs\_spec' NCS specification file with all NCS information `params_out` = `find_ncs_from_density.params.eff` Parameters file to rerun `find_ncs_from_density` `ncs_mask_file` = None If defined, map coefficients will be written to this file that show the asymmetric unit of NCS. `directory` `temp_dir` = "temp\_dir" Temporary work directory `output_dir` = "" Output directory where files are to be written `density_search` `density_radius` = 10. Radius for density to be cut out and compared `peak_separation` = 15 Minimum distance between centers. Use about 1.5\*`density_radius` `density_peaks` = 20 Number of NCS-related peaks of density to output `delta_phi` = 20 Angular spacing of search `rotz_only` = False Rotate around Z only `ncs_copies_max` = None Maximum number of NCS copies to look for `min_ratio_to_top_cc` = 0.75 Peaks will be kept up to `ncs_copies_max`, or `min_ratio_to_top_cc` \* best cc, whichever comes first `minimum_ncs_cc_from_density` = 0.30 Overall NCS CC at full resolution must be at least this high `fraction_ncs_min_from_density` = 0.05 Minimum fraction of the asymmetric unit covered by NCS. NCS ignored if less than this. `dump_ncs_density` = False You can dump 1 mtz file for each possible NCS copy so that you can see if the NCS is real Each file has the local map transformed to the orientation and position of copy 1 so they should all superimpose The files are written to the `temp_dir` (e.g., `temp_dir/ncs_1_map_coeffs_001.mtz`) `map_operators_inside_unit_cell` = False If set then put the operators inside the unit cell. This removes the use of space-group symmetry for SG `P1` and is used normally for non-crystal data. `find_centers` `smoothing_radius` = None Radius for smoothing squared density to find centers Choose a smaller value to get more center guesses. Default is same value as `density_radius`. `n_center_find` = None Target number of centers to find Smoothing radius will be varied from 2 to 2\*`smoothing_radius` and the value giving the number of peaks closest to `n_center_find` will be used. Normally leave this set to None `n_center_use` = 20 Number of locations to consider as molecular centers in density search. Try increasing this number to look harder for NCS. `try_all_centers` = False If set, try all `n_center_use` centers. Normally stop if a good solution is found. `ok_ncs_cc` = 0.6 NCS correlation good enough to terminate search `crystal_info` `resolution` = 4. High-resolution limit for map calculation It is useful to cut the resolution at 3-5 Å If you use higher resolution, use a finer `delta_phi` solvent fraction `control` verbose = True Verbose output `debug` = False Debugging output `raise_sorry` = False Raise sorry if problems `dry_run` = False Just read in and check parameter names `resolve_command_list` = None You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (&quot;,) and each individual command in single quotes (') like this: `resolve_command_list=&quot;'no_build...`

- `input_filesmtz_in = None` MTZ file with coefficients for a map `labin = ""` Labin line for MTZ file with map coefficients. This is optional if `find_ncs_from_density` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where `myFP` is your column label for FP `center_pdb_in = None` Optional PDB file with list of centers of density to be used as search model `density_mtz_in = None` MTZ file with coefficients for density around center defined by first atom in `center_pdb_in` (required) NOTE: Must be in space group p1 with identical cell to `mtz_in` `density_labin = ""` Labin line for MTZ file `density_mtz_in`. This is optional if `find_ncs_from_density` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where `myFP` is your column label for FP
- `mtz_in = None` MTZ file with coefficients for a map
- `labin = ""` Labin line for MTZ file with map coefficients. This is optional if `find_ncs_from_density` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where `myFP` is your column label for FP
- `center_pdb_in = None` Optional PDB file with list of centers of density to be used as search model
- `density_mtz_in = None` MTZ file with coefficients for density around center defined by first atom in `center_pdb_in` (required) NOTE: Must be in space group p1 with identical cell to `mtz_in`
- `density_labin = ""` Labin line for MTZ file `density_mtz_in`. This is optional if `find_ncs_from_density` can guess the correct coefficients for FP PHI and FOM. Otherwise specify: `LABIN FP=myFP PHIB=myPHI FOM=myFOM` where `myFP` is your column label for FP
- `output_filescenters_pdb_out = guess_molecular_centers.pdb` Output PDB file with coords of centers `log = find_ncs_from_density.log` Output log file `ncs_spec_file = 'find_ncs_from_density.ncs_spec'` NCS specification file with all NCS information `params_out = find_ncs_from_density_params.eff` Parameters file to rerun `find_ncs_from_density` `ncs_mask_file = None` If defined, map coefficients will be written to this file that show the asymmetric unit of NCS.
- `centers_pdb_out = guess_molecular_centers.pdb` Output PDB file with coords of centers
- `log = find_ncs_from_density.log` Output log file
- `ncs_spec_file = 'find_ncs_from_density.ncs_spec'` NCS specification file with all NCS information
- `params_out = find_ncs_from_density_params.eff` Parameters file to rerun `find_ncs_from_density`
- `ncs_mask_file = None` If defined, map coefficients will be written to this file that show the asymmetric unit of NCS.
- `temp_dir = "temp_dir"` Temporary work directory `output_dir = ""` Output directory where files are to be written
- `temp_dir = "temp_dir"` Temporary work directory
- `output_dir = ""` Output directory where files are to be written
- `density_searchdensity_radius = 10` Radius for density to be cut out and compared `peak_separation = 15` Minimum distance between centers. Use about  $1.5 \times \text{density\_radius}$  `density_peaks = 20` Number of NCS-related peaks of density to output `delta_phi = 20` Angular spacing of search `rotz_only = False` Rotate around Z only `ncs_copies_max = None` Maximum number of NCS copies to look for `min_ratio_to_top_cc = 0.75` Peaks will be kept up to `ncs_copies_max`, or `min_ratio_to_top_cc * best cc`, whichever comes first `minimum_ncs_cc_from_density = 0.30` Overall NCS CC at full resolution must be at least this high `fraction_ncs_min_from_density = 0.05` Minimum fraction of the asymmetric unit covered by NCS. NCS ignored if less than this. `dump_ncs_density = False` You can dump 1 mtz file for each possible NCS copy so that you can see if the NCS is real Each file has the local map transformed to the orientation and position of copy 1 so they should all superimpose The files are written to the `temp_dir` (e.g.,

temp\_dir/ncs\_1\_map\_coeffs\_001.mtz )map\_operators\_inside\_unit\_cell = False If set then put the operators inside the unit cell. This removes the use of space-group symmetry for SG P1 and is used normally for non-crystal data.

- density\_radius = 10. Radius for density to be cut out and compared
- peak\_separation = 15 Minimum distance between centers. Use about 1.5\*density\_radius
- density\_peaks = 20 Number of NCS-related peaks of density to output
- delta\_phi = 20 Angular spacing of search
- rotz\_only = False Rotate around Z only
- ncs\_copies\_max = None Maximum number of NCS copies to look for
- min\_ratio\_to\_top\_cc = 0.75 Peaks will be kept up to ncs\_copies\_max, or min\_ratio\_to\_top\_cc \* best cc, whichever comes first
- minimum\_ncs\_cc\_from\_density = 0.30 Overall NCS CC at full resolution must be at least this high
- fraction\_ncs\_min\_from\_density = 0.05 Minimum fraction of the asymmetric unit covered by NCS. NCS ignored if less than this.
- dump\_ncs\_density = False You can dump 1 mtz file for each possible NCS copy so that you can see if the NCS is real Each file has the local map transformed to the orientation and position of copy 1 so they should all superimpose The files are written to the temp\_dir (e.g., temp\_dir/ncs\_1\_map\_coeffs\_001.mtz )
- map\_operators\_inside\_unit\_cell = False If set then put the operators inside the unit cell. This removes the use of space-group symmetry for SG P1 and is used normally for non-crystal data.
- find\_centerssmoothing\_radius = None Radius for smoothing squared density to find centers Choose a smaller value to get more center guesses. Default is same value as density\_radius.n\_center\_find = None Target number of centers to find Smoothing radius will be varied from 2 to 2\*smoothing\_radius and the value giving the number of peaks closest to n\_center\_find will be used. Normally leave this set to None
- n\_center\_use = 20 Number of locations to consider as molecular centers in density search. Try increasing this number to look harder for NCS.try\_all\_centers = False If set, try all n\_center\_use centers. Normally stop if a good solution is found.ok\_ncs\_cc = 0.6 NCS correlation good enough to terminate search
- smoothing\_radius = None Radius for smoothing squared density to find centers Choose a smaller value to get more center guesses. Default is same value as density\_radius.
- n\_center\_find = None Target number of centers to find Smoothing radius will be varied from 2 to 2\*smoothing\_radius and the value giving the number of peaks closest to n\_center\_find will be used. Normally leave this set to None
- n\_center\_use = 20 Number of locations to consider as molecular centers in density search. Try increasing this number to look harder for NCS.
- try\_all\_centers = False If set, try all n\_center\_use centers. Normally stop if a good solution is found.
- ok\_ncs\_cc = 0.6 NCS correlation good enough to terminate search
- crystal\_inforesolution = 4. High-resolution limit for map calculation It is useful to cut the resolution at 3-5 A If you use higher resolution, use a finer delta\_phisolvent\_fraction = 0.5 solvent fraction
- resolution = 4. High-resolution limit for map calculation It is useful to cut the resolution at 3-5 A If you use higher resolution, use a finer delta\_phi
- solvent\_fraction = 0.5 solvent fraction

- `controlverbose = True` Verbose output
- `debug = False` Debugging output
- `raise_sorry = False` Raise sorry if problems
- `dry_run = False` Just read in and check parameter names
- `resolve_command_list = None` You can supply any resolve command here NOTE: for command-line usage you need to enclose the whole set of commands in double quotes (&quot;) and each individual command in single quotes (') like this: `resolve_command_list=&quot;'no_build' 'b_overall 23' &quot;`

# phenix.map\_box: extract box with model and map around selected atoms

---

map\_box.html

## Description

The MapBox program is normally used with cryoEM data. MapBox carves out a box with model and map around selected atoms. The output map is produced in three ways: map coefficients in MTZ, and maps in CCP4 and X-plor formatted files.

## Contact author

For questions, bug reports, feature requests: Pavel Afonine (PAfonine@lbl.gov)

## Usage examples

Using a map defined in MTZ file as Fourier map coefficients. Box is defined by selected atoms in residues from 1 to 10 in chain A `phenix.map_box model.pdb map_coefficients.mtz selection="chain A and resseq 1:10"`  
Using a CCP4/mrc formatted map. `phenix.map_box model.pdb map.ccp4 selection="resname LIG"` Note that by default map\_box chooses a rectangular box that is a little bigger than your model. If you want the map to be zeroed out away from your model you can use the keyword `mask_atoms=True`. Extract unique part of a map. You can extract just the unique part of a map (unique based on map symmetry). This map is masked around high density. It will normally only have positive density and lower density parts of even the unique part are truncated. Note that if you want to sharpen the map, you must sharpen it before you extract the unique part. You should not sharpen a map file created with `extract_unique` because the `extract_unique` function creates a map that has been set to zero over much of the map and this interferes with auto-sharpen.

`phenix.map_box map.ccp4 extract_unique=True symmetry.ncs_spec resolution=3` Shift origin of a map. You can shift the origin of a map (redefine where in the unit\_cell box the map goes). Normally you need to also specify the the map size should remain the same. You can specify the origin directly: `phenix.map_box map.ccp4 keep_map_size=True prefix=map_origin_shifted output_origin_grid_units=3,5,7` You can also specify the origin to be (0,0,0) with `keep_origin=False`: `phenix.map_box map.ccp4 keep_map_size=True prefix=map_origin_shifted keep_origin=False` You can also specify that the origin should match the origin of some other file. If you do this you have to use the keyword `ccp4_map_file` for the map to be shifted and `output_origin_match_this_file` for the one to match: `phenix.map_box ccp4_map_file=map.ccp4`

`keep_map_size=True prefix=map_origin_shifted`

`output_origin_match_this_file=map_with_origin_to_match.ccp4` Cut out a part of a map to match some other map. You can say: `phenix.map_box ccp4_map_file=map.ccp4 prefix=map_boxed`

`bounds_match_this_file=map_with_bounds_to_match.ccp4` Invert the hand of a map: `phenix.map_box ccp4_map_file=map.ccp4 keep_map_size=True invert_hand = True`

- Using a map defined in MTZ file as Fourier map coefficients. Box is defined by selected atoms in residues from 1 to 10 in chain A `phenix.map_box model.pdb map_coefficients.mtz selection="chain A and resseq 1:10"`

Using a map defined in MTZ file as Fourier map coefficients. Box is defined by selected atoms in residues from 1 to 10 in chain A

`phenix.map_box model.pdb map_coefficients.mtz selection="chain A and resseq 1:10"`

- Using a CCP4/mrc formatted map. `phenix.map_box model.pdb map.ccp4 selection="resname LIG"`  
Note that by default map\_box chooses a rectangular box that is a little bigger than your model. If you

want the map to be zeroed out away from your model you can use the keyword `mask_atoms=True`.

Using a CCP4/mrc formatted map.

```
phenix.map_box model.pdb map.ccp4 selection="resname LIG"
```

Note that by default `map_box` chooses a rectangular box that is a little bigger than your model. If you want the map to be zeroed out away from your model you can use the keyword `mask_atoms=True`.

- Extract unique part of a map. You can extract just the unique part of a map (unique based on map symmetry). This map is masked around high density. It will normally only have positive density and lower density parts of even the unique part are truncated. Note that if you want to sharpen the map, you must sharpen it before you extract the unique part. You should not sharpen a map file created with `extract_unique` because the `extract_unique` function creates a map that has been set to zero over much of the map and this interferes with auto-sharpen. `phenix.map_box map.ccp4 extract_unique=True symmetry.ncs_spec resolution=3`

Extract unique part of a map. You can extract just the unique part of a map (unique based on map symmetry). This map is masked around high density. It will normally only have positive density and lower density parts of even the unique part are truncated. Note that if you want to sharpen the map, you must sharpen it before you extract the unique part. You should not sharpen a map file created with `extract_unique` because the `extract_unique` function creates a map that has been set to zero over much of the map and this interferes with auto-sharpen.

```
phenix.map_box map.ccp4 extract_unique=True symmetry.ncs_spec resolution=3
```

- Shift origin of a map. You can shift the origin of a map (redefine where in the unit\_cell box the map goes). Normally you need to also specify the the map size should remain the same. You can specify the origin directly: `phenix.map_box map.ccp4 keep_map_size=True prefix=map_origin_shifted output_origin_grid_units=3,5,7` You can also specify the origin to be (0,0,0) with `keep_origin=False`: `phenix.map_box map.ccp4 keep_map_size=True prefix=map_origin_shifted keep_origin=False` You can also specify that the origin should match the origin of some other file. If you do this you have to use the keyword `ccp4_map_file` for the map to be shifted and `output_origin_match_this_file` for the one to match: `phenix.map_box ccp4_map_file=map.ccp4 keep_map_size=True prefix=map_origin_shifted output_origin_match_this_file=map_with_origin_to_match.ccp4`

Shift origin of a map. You can shift the origin of a map (redefine where in the unit\_cell box the map goes). Normally you need to also specify the the map size should remain the same.

You can specify the origin directly:

```
phenix.map_box map.ccp4 keep_map_size=True prefix=map_origin_shifted
output_origin_grid_units=3,5,7
output_origin_grid_units=3,5,7
```

You can also specify the origin to be (0,0,0) with `keep_origin=False`:

```
phenix.map_box map.ccp4 keep_map_size=True prefix=map_origin_shifted
keep_origin=False
keep_origin=False
```

You can also specify that the origin should match the origin of some other file. If you do this you have to use the keyword `ccp4_map_file` for the map to be shifted and `output_origin_match_this_file` for the one to match:

```
phenix.map_box ccp4_map_file=map.ccp4 keep_map_size=True
```

prefix=map\_origin\_shifted output\_origin\_match\_this\_file=map\_with\_origin\_to\_match.ccp4

prefix=map\_origin\_shifted output\_origin\_match\_this\_file=map\_with\_origin\_to\_match.ccp4

- Cut out a part of a map to match some other map. You can say: phenix.map\_box  
ccp4\_map\_file=map.ccp4 prefix=map\_boxed bounds\_match\_this\_file=map\_with\_bounds\_to\_match.ccp4

Cut out a part of a map to match some other map. You can say:

phenix.map\_box ccp4\_map\_file=map.ccp4

prefix=map\_boxed bounds\_match\_this\_file=map\_with\_bounds\_to\_match.ccp4

prefix=map\_boxed bounds\_match\_this\_file=map\_with\_bounds\_to\_match.ccp4

- Invert the hand of a map: phenix.map\_box ccp4\_map\_file=map.ccp4 keep\_map\_size=True invert\_hand = True

Invert the hand of a map:

phenix.map\_box ccp4\_map\_file=map.ccp4

keep\_map\_size=True invert\_hand = True

keep\_map\_size=True invert\_hand = True

## Notes

By default the output map will superimpose on the input map (keep\_origin=True). The output map is normally smaller than the input map. For example, the input map might go from (0,0,0) to (100,100,100) and the output map might include only grid points from (50,50,50) to (100,100,100). The origin of the output map in this case will be (50,50,50). If the two maps are displayed, the region from (50,50,50) to (100,100,100) will be the same for the two maps. Not that anything outside of this range is meaningless for the output map. The cell dimensions of the output map will be the dimensions of the part that is cut out. In the above example, if the starting unit cell was (200,200,200) then the output map will have a unit cell of (100,100,100). Some maps may have a special tag called external\_origin. In the map file it is called ORIGIN. This is different than the origin of the map, which specifies the grid point that corresponds to the first element in the map. The external\_origin is a coordinate shift that may correspond to this grid point, but that does not have to fall right on a grid point. If a map (1) has an external\_origin tag and (2) the external\_origin falls on a grid point in the map, and (3) the map does not have an origin specified in the usual way, then the external\_origin, converted to grid units, will be used as the origin of the map in Phenix. You can use phenix.show\_map\_info to display characteristics of your map. If you want to remove the external\_origin tag from your map, you can run map\_box with "keep\_map\_size=True" "keep\_origin=False" and it will write out the same map but without the external\_origin tag and without an origin shift. If you want to convert your external\_origin tag to an origin shift, you can run map\_box with "keep\_map\_size=True" and "keep\_origin=True". Also, if you want to shift a model to match a map with an external\_origin tag, you can use the tool phenix.shift\_model\_to\_match\_map (command-line only). How map gridding is described in mrc/ccp4 format. Map specifications. A full map is defined for a full box or unit cell, however in many cases only a part of that map is present. The full map is on a grid of 'unit cell grid' which corresponds to one unit cell (box) with dimensions of 'unit cell parameters'. The map that is present is on part of the 'unit cell grid'. It starts at the grid point called the 'origin' and goes to one grid point in each direction before the grid point called 'last'. The map that is present has 'all' grid points in each direction. The map that is present has a 'map unit cell' and 'map grid' that correspond to the dimensions 'all'. Example: a full map with a unit cell grid of (10, 10, 10) goes from (0, 0, 0) to (9, 9, 9) and has dimensions of (10, 10, 10). Example: a partial map with a unit cell grid of (10,10,10) and an origin of (0, 0, 2) and a map grid or 'all' of (6, 6, 6), would go from (0, 0, 2) to (5, 5, 7). Ignoring symmetry conflicts. If you cut out a part of a map, then the new map will have a box size that is different from the original one. If that box size does not match the box size

for your PDB file or other map, then you may get an error message that says the symmetry of the maps or map and model is not the same. You can ignore this with the keyword `ignore_symmetry_conflicts=True`. Running `map_box` with bounds that are outside the map. You can specify any lower and upper bounds for your output map, but if the bounds are outside the map you supply the values may not make any sense. By default the map you supply is wrapped so that the value for a point outside the map is taken from the grid point that would be in this place if another copy of the map were placed right next to your map. You can turn this off and instead supply zeroes outside the map with the keyword `restrict_map_size=True`. What to do after MapBox. Normally you will have cut out part of the map with MapBox for the purpose of working with just part of the map. You might carry out automatic mode...

- By default the output map will superimpose on the input map (`keep_origin=True`). The output map is normally smaller than the input map. For example, the input map might go from (0,0,0) to (100,100,100) and the output map might include only grid points from (50,50,50) to (100,100,100). The origin of the output map in this case will be (50,50,50). If the two maps are displayed, the region from (50,50,50) to (100,100,100) will be the same for the two maps. Not that anything outside of this range is meaningless for the output map.

By default the output map will superimpose on the input map (`keep_origin=True`). The output map is normally smaller than the input map. For example, the input map might go from (0,0,0) to (100,100,100) and the output map might include only grid points from (50,50,50) to (100,100,100). The origin of the output map in this case will be (50,50,50). If the two maps are displayed, the region from (50,50,50) to (100,100,100) will be the same for the two maps. Not that anything outside of this range is meaningless for the output map.

- The cell dimensions of the output map will be the dimensions of the part that is cut out. In the above example, if the starting unit cell was (200,200,200) then the output map will have a unit cell of (100,100,100).

The cell dimensions of the output map will be the dimensions of the part that is cut out. In the above example, if the starting unit cell was (200,200,200) then the output map will have a unit cell of (100,100,100).

- Some maps may have a special tag called `external_origin`. In the map file it is called `ORIGIN`. This is different than the origin of the map, which specifies the grid point that corresponds to the first element in the map. The `external_origin` is a coordinate shift that may correspond to this grid point, but that does not have to fall right on a grid point. If a map (1) has an `external_origin` tag and (2) the `external_origin` falls on a grid point in the map, and (3) the map does not have an origin specified in the usual way, then the `external_origin`, converted to grid units, will be used as the origin of the map in Phenix. You can use `phenix.show_map_info` to display characteristics of your map. If you want to remove the `external_origin` tag from your map, you can run `map_box` with "`keep_map_size=True`" "`keep_origin=False`" and it will write out the same map but without the `external_origin` tag and without an origin shift. If you want to convert your `external_origin` tag to an origin shift, you can run `map_box` with "`keep_map_size=True`" and "`keep_origin=True`". Also, if you want to shift a model to match a map with an `external_origin` tag, you can use the tool `phenix.shift_model_to_match_map` (command-line only).

Some maps may have a special tag called `external_origin`. In the map file it is called `ORIGIN`. This is different than the origin of the map, which specifies the grid point that corresponds to the first element in the map. The `external_origin` is a coordinate shift that may correspond to this grid point, but that does not have to fall right on a grid point.

If a map (1) has an `external_origin` tag and (2) the `external_origin` falls on a grid point in the map, and (3) the map does not have an origin specified in the usual way, then the `external_origin`, converted to grid units, will be used as the origin of the map in Phenix.

You can use `phenix.show_map_info` to display characteristics of your map. If you want to remove the `external_origin` tag from your map, you can run `map_box` with `"keep_map_size=True"` `"keep_origin=False"` and it will write out the same map but without the `external_origin` tag and without an origin shift. If you want to convert your `external_origin` tag to an origin shift, you can run `map_box` with `"keep_map_size=True"` and `"keep_origin=True"`. Also, if you want to shift a model to match a map with an `external_origin` tag, you can use the tool `phenix.shift_model_to_match_map` (command-line only).

- How map gridding is described in mrc/ccp4 format. Map specifications. A full map is defined for a full box or unit cell, however in many cases only a part of that map is present. The full map is on a grid of 'unit cell grid' which corresponds to one unit cell (box) with dimensions of 'unit cell parameters'. The map that is present is on part of the 'unit cell grid'. It starts at the grid point called the 'origin' and goes to one grid point in each direction before the grid point called 'last'. The map that is present has 'all' grid points in each direction. The map that is present has a 'map unit cell' and 'map grid' that correspond to the dimensions 'all'. Example: a full map with a unit cell grid of (10, 10, 10) goes from (0, 0, 0) to (9, 9, 9) and has dimensions of (10, 10, 10). Example: a partial map with a unit cell grid of (10,10,10) and an origin of (0, 0, 2) and a map grid or 'all' of (6, 6, 6), would go from (0, 0, 2) to (5, 5, 7).

How map gridding is described in mrc/ccp4 format.

Map specifications. A full map is defined for a full box or unit cell, however in many cases only a part of that map is present. The full map is on a grid of 'unit cell grid' which corresponds to one unit cell (box) with dimensions of 'unit cell parameters'.

The map that is present is on part of the 'unit cell grid'. It starts at the grid point called the 'origin' and goes to one grid point in each direction before the grid point called 'last'. The map that is present has 'all' grid points in each direction. The map that is present has a 'map unit cell' and 'map grid' that correspond to the dimensions 'all'.

Example: a full map with a unit cell grid of (10, 10, 10) goes from (0, 0, 0) to (9, 9, 9) and has dimensions of (10, 10, 10).

Example: a partial map with a unit cell grid of (10,10,10) and an origin of (0, 0, 2) and a map grid or 'all' of (6, 6, 6), would go from (0, 0, 2) to (5, 5, 7).

- Ignoring symmetry conflicts. If you cut out a part of a map, then the new map will have a box size that is different from the original one. If that box size does not match the box size for your PDB file or other map, then you may get an error message that says the symmetry of the maps or map and model is not the same. You can ignore this with the keyword `ignore_symmetry_conflicts=True`.

Ignoring symmetry conflicts. If you cut out a part of a map, then the new map will have a box size that is different from the original one. If that box size does not match the box size for your PDB file or other map, then you may get an error message that says the symmetry of the maps or map and model is not the same. You can ignore this with the keyword `ignore_symmetry_conflicts=True`.

- Running `map_box` with bounds that are outside the map. You can specify any lower and upper bounds for your output map, but if the bounds are outside the map you supply the values may not make any sense. By default the map you supply is wrapped so that the value for a point outside the map is taken from the grid point that would be in this place if another copy of the map were placed right next to your map. You can turn this off and instead supply zeroes outside the map with the keyword `restrict_map_size=True`.

Running `map_box` with bounds that are outside the map. You can specify any lower and upper bounds for your output map, but if the bounds are outside the map you supply the values may not make any sense. By default the map you supply is wrapped so that the value for a point outside the map is taken from the grid

point that would be in this place if another copy of the map were placed right next to your map. You can turn this off and instead supply zeroes outside the map with the keyword `restrict_map_size=True`.

- What to do after MapBox. Normally you will have cut out part of the map with MapBox for the purpose of working with just part of the map. You might carry out automatic model-building with `map_to_model` or with `predict_and_build`, or real-space refinement, or manual model-building with Coot or Isolve after MapBox.

What to do after MapBox. Normally you will have cut out part of the map with MapBox for the purpose of working with just part of the map. You might carry out automatic model-building with `map_to_model` or with `predict_and_build`, or real-space refinement, or manual model-building with Coot or Isolve after MapBox.

#### List of all parameters

`job_title` = None Job title in PHENIX GUI, not used on command line  
`pdbe_file` = None Optional model file used to define region to be cut out  
`map_coefficients_file` = None Input map coefficients file (alternative to ccp4 map file)  
`label` = None Labels for map coefficients file  
`ccp4_map_file` = None Input map file (CCP4/mrc format)  
`mask_file_name` = None Input mask file (CCP4/mrc format)  
`target_ncs_au_file` = None File with model indicating which au to choose in  
`extract_uniqueselection` = all Atom selection to be applied to input PDB file  
`selection_radius` = 3.0 Atoms within `selection_radius` of a selected atom model will be kept as part of the selection.  
`box_cushion` = 3.0 If model is supplied, a box of density will be cut out around the input model (after selections are applied to the model). The size of the box of density will be `box_cushion` bigger than the model. Box cushion also applied if `density_select` is set. Box cushion is also used in `extract_unique`.  
`mask_atoms` = False Set map values to 0 outside molecular mask  
`mask_atoms_atom_radius` = 3. Radius for masking around atoms  
`write_mask_file` = False Write mask file. Requires setting `mask_atomsset_outside_to_mean_inside` = False Set value outside mask equal to mean inside  
`resolution_factor` = 1./4 Resolution factor for calculation of map coefficients  
`map_scale_factor` = None Scale factor to apply to map  
`scale_max` = 99999 Maximum value of amplitudes for output mtz file. If None, apply volume scaling  
`resolution` = None Resolution for calculation of output map coefficients. Default is based on the gridding of the map (and may be higher-resolution than you want).  
`output_format` = xplor \*mtz \*ccp4 Output format(s) for boxed map. Note that mtz format is only available if `keep_origin` = False or `keep_map_size` = True. (These are the cases where the map is cut down to size and placed at the origin or there is a full unit cell of data.)  
`output_file_name_prefix` = None Prefix for output file names. Default is name of the pdb file without the .pdb suffix.  
`mask_select` = False Select boundaries (min, max in x, y, z) based on auto-mask  
`density_select` = False Select boundaries based on where density is located.  
`density_select_threshold` = 0.05 Choose region where density is this fraction of maximum or greater  
`get_half_height_width` = True Use 4 times half-width at half-height as estimate of max size in density  
`selectsymmetry` = None Optional symmetry (e.g., D7, I, C2) to be used if `extract_unique` is set. Alternative to `symmetry_file`. To find symmetry automatically specify `symmetry` = ALL.  
`symmetry_file` = None Symmetry file. Symmetry or `symmetry_file` required if `extract_unique` = True. May be a Phenix .ncs\_spec file or BIOMTR records or a resolve ncs file.  
`sequence_file` = None Sequence file (any standard format). Can be unique part or all copies. Used in identification of unique part of map and in masking with  
`mask_selectmolecular_mass` = None Molecular mass of object in map in Da (i.e., 33000 for 33 Kd). Used in identification of unique part of map and in masking by `mask_select`.  
`solvent_content` = None Optional fraction of volume of map that is empty. Used in identification of unique part of map and in masking by  
`mask_selectextract_unique` = False Extract unique part of map. Requires either `symmetry_file` or `symmetry` and either sequence file or molecular mass to be supplied. If chain type is not protein it should be set as well.  
`use_cubic_boxing` = None When boxing map, create a box that is cubic (same number of grid points in each direction). Can be used with `stay_inside_current_map` to keep the box inside the current map. Note that if box is bigger than the current map and wrapping is False, any points outside original map will be set to zero. Can be used with `extract_unique`, `mask_select`, `density_select` and default boxing around molecule.

Incompatible with lower\_bounds and upper\_bounds. stay\_inside\_current\_map = None If True, then when applying boxing, keep the new box inside the current map. If False, allow new box to go outside the current...

- job\_title = None Job title in PHENIX GUI, not used on command line
- pdb\_file = None Optional model file used to define region to be cut out
- map\_coefficients\_file = None Input map coefficients file (alternative to ccp4 map file)
- label = None Labels for map coefficients file
- ccp4\_map\_file = None Input map file (CCP4/mrc format).
- mask\_file\_name = None Input mask file (CCP4/mrc format).
- target\_ncs\_au\_file = None File with model indicating which au to choose in extract\_unique
- selection = all Atom selection to be applied to input PDB file
- selection\_radius = 3.0 Atoms within selection\_radius of a selected atom model will be kept as part of the selection.
- box\_cushion = 3.0 If model is supplied, a box of density will be cut out around the input model (after selections are applied to the model). The size of the box of density will be box\_cushion bigger than the model. Box cushion also applied if density\_select is set. Box cushion is also used in extract\_unique.
- mask\_atoms = False Set map values to 0 outside molecular mask
- mask\_atoms\_atom\_radius = 3. Radius for masking around atoms
- write\_mask\_file = False Write mask file. Requires setting mask\_atoms
- set\_outside\_to\_mean\_inside = False Set value outside mask equal to mean inside
- resolution\_factor = 1./4 Resolution factor for calculation of map coefficients
- map\_scale\_factor = None Scale factor to apply to map
- scale\_max = 99999 Maximum value of amplitudes for output mtz file. If None, apply volume scaling
- resolution = None Resolution for calculation of output map coefficients. Default is based on the gridding of the map (and may be higher-resolution than you want).
- output\_format = xplor \*mtz \*ccp4 Output format(s) for boxed map. Note that mtz format is only available if keep\_origin = False or keep\_map\_size = True. (These are the cases where the map is cut down to size and placed at the origin or there is a full unit cell of data.)
- output\_file\_name\_prefix = None Prefix for output file names. Default is name of the pdb file without the .pdb suffix.
- mask\_select = False Select boundaries (min, max in x, y, z) based on auto-mask
- density\_select = False Select boundaries based on where density is located.
- density\_select\_threshold = 0.05 Choose region where density is this fraction of maximum or greater
- get\_half\_height\_width = True Use 4 times half-width at half-height as estimate of max size in density\_select
- symmetry = None Optional symmetry (e.g., D7, I, C2) to be used if extract\_unique is set. Alternative to symmetry\_file. To find symmetry automatically specify symmetry = ALL.
- symmetry\_file = None Symmetry file. Symmetry or symmetry\_file required if extract\_unique = True. May be a Phenix .ncs\_spec file or BIOMTR records or a resolve ncs file.
- sequence\_file = None Sequence file (any standard format). Can be unique part or all copies. Used in identification of unique part of map and in masking with mask\_select

- `molecular_mass` = None Molecular mass of object in map in Da (i.e., 33000 for 33 Kd). Used in identification of unique part of map and in masking by `mask_select`.
- `solvent_content` = None Optional fraction of volume of map that is empty. Used in identification of unique part of map and in masking by `mask_select`
- `extract_unique` = False Extract unique part of map. Requires either `symmetry_file` or `symmetry` and either sequence file or molecular mass to be supplied. If chain type is not protein it should be set as well.
- `use_cubic_boxing` = None When boxing map, create a box that is cubic (same number of grid points in each direction). Can be used with `stay_inside_current_map` to keep the box inside the current map. Note that if box is bigger than the current map and `wrapping` is False, any points outside original map will be set to zero. Can be used with `extract_unique`, `mask_select`, `density_select` and default boxing around molecule. Incompatible with `lower_bounds` and `upper_bounds`.
- `stay_inside_current_map` = None If True, then when applying boxing, keep the new box inside the current map. If False, allow new box to go outside the current map. Values outside will be zero if `wrapping=False` as determined from input map or the parameter `wrapping`, and wrapped values if `wrapping=True`
- `increase_box_cushion_and_atom_radius_for_soft_mask` = True Expand cushion and atom radii by `soft_mask_radius`
- `soft_mask_extract_unique` = True Create soft mask at edges of `extract_unique` box (feather map into edge of box). Uses resolution as `mask_radius`
- `mask_expand_ratio` = 1 Mask expansion relative to resolution for `extract_unique`
- `regions_to_keep` = None You can specify a limit to the number of regions to keep when generating the asymmetric unit of density.
- `keep_low_density` = True Get remainder (weak density) with `extract_unique`.
- `chain_type` = None \*PROTEIN DNA RNA Chain type. Only used if `extract_unique` is set. Has minor effect in setting thresholds for identification of molecular region. Use None if there is a mixture.
- `soft_mask` = False Use Gaussian mask in `mask_atoms` and on outside surface of box
- `invert_mask` = False Make mask with 1 outside model (only applies to `mask_around_atoms`)
- `soft_mask_radius` = 3 Gaussian mask smoothing radius
- `lower_bounds` = None Lower bounds for cut out box. You can specify them directly. NOTE: lower and upper bounds refer to grid points after shifting the map to place the origin at (0, 0, 0). To refer to absolute values specify `bounds_are_absolute` = True.
- `upper_bounds` = None Upper bounds for cut out box. You can specify them directly. NOTE: lower and upper bounds refer to grid points after shifting the map to place the origin at (0, 0, 0). To refer to absolute values specify `bounds_are_absolute` = True.
- `bounds_are_absolute` = False Define lower and upper bounds as absolute. NOTE: lower and upper bounds refer to grid points after shifting the map to place the origin at (0, 0, 0). To refer to absolute values specify `bounds_are_absolute` = True.
- `keep_map_size` = False Keep original map gridding (do not cut anything out). Use to apply `soft_mask` and/or `mask_atoms` keeping same map size.
- `keep_origin` = True Write out map, `map_coefficients`, and model with origin in original location. If False, shift the origin to (0, 0, 0). NOTE: to cut out a part of a map, shift the origin to (0, 0, 0), and make a new small map use `keep_origin` = False `keep_input_unit_cell_and_grid` = False

- `keep_input_unit_cell_and_grid = True` Keep the input `unit_cell` dimensions and `unit_cell_grid`. If `False`, use the dimensions and grid of the cut out box as the unit cell `map_box` dimensions and grid. NOTE: to cut out a part of a map, shift the origin to (0, 0, 0), and make a new small map set `keep_origin = False` and `keep_input_unit_cell_and_grid = False`
- `output_unit_cell = None` You can specify the unit cell for your map with 3 numbers. This should normally not be necessary. It can be used to fix a map that has the wrong unit cell.
- `output_unit_cell_grid = None` You can specify the grid (3 integers) corresponding to the output unit cell. This can be used to specify the full grid for the unit cell. if `output_unit_cell` is not specified, new unit cell parameters will be generated to maintain the grid spacing.
- `output_origin_grid_units = None` You can specify the origin of your output map. Normally you should use `keep_origin = True` or `False` to specify your origin but if you want to move it to a specific grid point you can do that. `.short_caption = Output origin`
- `output_origin_match_this_file = None` As `output_origin_grid_units`, but use origin from this file
- `bounds_match_this_file = None` Take the lower and upper bounds from this map file and apply them to the input map file.
- `output_external_origin = None` Write ORIGIN record to map file (this is an external origin used to specify relationship to external files such as model files. Three floating point numbers (A).
- `restrict_map_size = False` Do not go outside original map boundaries
- `ignore_symmetry_conflicts = False` Ignore unit cell from model if it conflicts with the map.
- `wrapping = False` If wrapping, map wraps around at map boundaries.
- `check_wrapping = False` Check that wrapping is consistent with map if it is set to `True`
- `output_ccp4_map_mean = None` Choose mean and SD of output CCP4 map
- `output_ccp4_map_sd = None` Choose mean and SD of output CCP4 map
- `output_map_label = None` Add this label to output map
- `remove_output_map_labels = None` Remove all output map labels
- `invert_hand = False` Just before writing out the map, swap the order of all sections in Z. This will change the hand of the map. Note that this removes any correspondence to models (these are not inverted). If you use this, be sure to apply it to all your starting maps. `.short_caption = Invert hand of map`
- guiGUI-specific parameter required for output directory `output_dir = None`
- `output_dir = None`

# Map comparison: Identify matching contour levels in two maps

---

map\_comparison.html

## Author(s)

- map\_comparison: Pavel Afonine

## Purpose

Identify matching contour levels in two maps

## How map\_comparison works:

## Examples

## Standard run of map\_comparison:

Running map\_comparison is easy. From the command-line you can type:

```
phenix.map_comparison map_1.mrc map_2.mrc
```

## Possible Problems

## Specific limitations and problems:

## Literature

Metrics for comparison of crystallographic maps. Urzhumtsev, A, Afonine, PV, Lunin, VY, Terwilliger, TC, Adams, and PD. Acta Crystallogr D Biol Crystallogr D70, 2593-2606 (2014).

- Metrics for comparison of crystallographic maps. Urzhumtsev, A, Afonine, PV, Lunin, VY, Terwilliger, TC, Adams, and PD. Acta Crystallogr D Biol Crystallogr D70, 2593-2606 (2014).

## Additional information

## List of all available keywords

job\_title = None Job title in PHENIX GUI, not used on command line  
inputmap\_1 = None A CCP4-formatted map  
map\_2 = None A CCP4-formatted map  
mapmtz\_1 = None MTZ file containing map  
mapmtz\_2 = None MTZ file containing map  
mapmtz\_label\_1 = None Data label for complex map coefficients in MTZ file  
mapmtz\_label\_2 = None Data label for complex map coefficients in MTZ file  
options.resolution\_factor = 0.25 Determines grid spacing in map  
shift\_origin = True Shift origin if necessary  
contour\_to\_match = None Contour level in map1 to match in map2 by volume equalization.

- job\_title = None Job title in PHENIX GUI, not used on command line
- inputmap\_1 = None A CCP4-formatted map  
map\_2 = None A CCP4-formatted map  
mapmtz\_1 = None MTZ file containing map  
mapmtz\_2 = None MTZ file containing map  
mapmtz\_label\_1 = None Data label for complex map coefficients in MTZ file  
mapmtz\_label\_2 = None Data label for complex map coefficients in MTZ file

- map\_1 = None A CCP4-formatted map
- map\_2 = None A CCP4-formatted map
- mtz\_1 = None MTZ file containing map
- mtz\_2 = None MTZ file containing map
- mtz\_label\_1 = None Data label for complex map coefficients in MTZ file
- mtz\_label\_2 = None Data label for complex map coefficients in MTZ file
- optionsresolution\_factor = 0.25 Determines grid spacing in mapshift\_origin = True Shift origin if necessarycontour\_to\_match = None Contour level in map1 to match in map2 by volume equalization.
- resolution\_factor = 0.25 Determines grid spacing in map
- shift\_origin = True Shift origin if necessary
- contour\_to\_match = None Contour level in map1 to match in map2 by volume equalization.

# Map correlations (map-model, map-map, optional origin shifts)

---

map\_correlations.html

## Author(s)

- map\_correlations: Pavel Afonine, Tom Terwilliger

## Purpose

Map correlations is a GUI and command-line tool for evaluating the fit of models to maps and the correlation between maps. It has the feature that the model or second map can be offset (translated) to maximize the correlation. This offset can be restricted to shifts allowed by space-group symmetry (translational\_offsets=Allowed\_by\_symmetry) or it can be any shifts (translational\_offsets=Any).

Maps can be supplied as map files or as map coefficients. Maps and map coefficients may not be both used at the same time.

If a model is supplied, the local correlation for each residue is calculated.

The total number of files supplied must be exactly two.

## Notes

The map\_correlations tool replaces phenix.map\_model\_cc, phenix.model\_map\_cc, phenix.get\_cc\_mtz\_mtz, phenix.get\_cc\_mtz\_pdb, and phenix.get\_cc\_map\_map.

## How map\_correlations works:

If an optimized translation is requested, map correlations finds the optimal translation using RESOLVE. The translation is calculated based on an FFT-based correlation search. If two maps (or map coefficients files) are supplied, the second map is shifted. If a model is supplied, a model-based map is calculated, and the optimal translation for that model-based map is applied to the model.

If you supply a model file, by default model density is calculated using the atom types, occupancies and isotropic thermal factors (B-values) supplied in that PDB file.

If you specify b\_values=Zero then all the B-values are set to zero before calculation of correlations. This is useful if you want to compare the clarities of a series of maps, all representing the same model. You calculate the map-model correlation of each map to your reference model in which all B-values in the reference model have been set to zero (Terwilliger et al., 2018):

## Output files from map\_correlations

xxx\_offset.pdb: A PDB file offset to match the origin in the mtz file. Produced if translational\_offsets are requested and a model is supplied. xxx\_offset.mtz: A translated version of the second MTZ file supplied (if any), offset so that the map calculated from it optimally matches the map from the first MTZ file.

xxx\_offset.ccp4: A translated version of the second map file supplied, (if any) offset so that the map calculated from it optimally matches the map from the first map file.

## Examples

## Standard run of map\_correlations:

Running map\_correlations is easy. From the command-line you can type:

```
phenix.map_correlations map_coeffs.mtz coords.pdb
```

## Possible Problems

### Specific limitations and problems:

If map and model are supplied, normally you must supply a value for the resolution to which map-model correlations are calculated.

## References

Automated map sharpening by maximization of detail and connectivity. T.C. Terwilliger, P.V. Afonine, Sobolev, OV, and P.D. Adams. Acta Cryst. D74, 545-559 (2018).

- Automated map sharpening by maximization of detail and connectivity. T.C. Terwilliger, P.V. Afonine, Sobolev, OV, and P.D. Adams. Acta Cryst. D74, 545-559 (2018).

## List of all available keywords

map\_correlationstranslational\_offsets = \*None Any Allowed\_by\_symmetry Allow origin shifts that are compatible with the map or map coefficients. For cryo-EM data, use None for no offsets or Any if you want to allow translations. For Xray data, use None for no offsets or Allowed\_by\_symmetry to allow origin offsets compatible with the space group.minimum\_cc\_for\_auto\_resolution = 0.1 Minimum CC for automatic resolution estimationb\_values = \*Keep Zero Keep B values as supplied, or set all to zero. Only applies if model is supplied.input\_filesmodel = None Modelmap\_in\_1 = None Map 1map\_in\_2 = None Map 2map\_coeffs\_1 = None Map coefficients 1map\_coeffs\_1\_virtual = None Used internallymap\_coeffs\_2\_virtual = None Used internallymap\_coeffs\_2 = None Map coefficients 2map\_coeffs\_labels\_1 = None map\_coeffs\_labels\_2 = None controlclean\_up = True Clean up temp directoriesverbose = False Verbose outputdirectoriestemp\_dir = None Optional temporary directorygui\_output\_dir = None GUI use only - does not apply to command line versionmap\_model\_ccresolution = None Data (map) resolutionscattering\_table = wk1995 it1992 n\_gaussian neutron \*electron Scattering table (X-ray, neutron or electron)atom\_radius = None Atom radius for masking. If undefined then calculated automaticallykeep\_map\_calc = False Keep model-calculated mapwrapping = None You can specify whether your maps wrap around outside boundariesignore\_symmetry\_conflicts = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with box\_map for example.computecc\_per\_chain = True Compute local model-map CC for each chaincc\_per\_residue = True Compute local model-map CC for each residuecc\_per\_residue\_group = False Compute local model-map CC for each residue groupfsc = True Compute FSCcc\_mask = True cc\_volume = True cc\_peaks = True cc\_box = True cc\_image = False outputfile\_name\_prefix = None Optional prefix (including path) for output files. Default is name of the program and run numberguiGUI-specific parameters, not used on command lineresult\_file = None job\_title = None Job title in PHENIX GUI, not used on command line

- map\_correlationstranslational\_offsets = \*None Any Allowed\_by\_symmetry Allow origin shifts that are compatible with the map or map coefficients. For cryo-EM data, use None for no offsets or Any if you want to allow translations. For Xray data, use None for no offsets or Allowed\_by\_symmetry to allow origin offsets compatible with the space group.minimum\_cc\_for\_auto\_resolution = 0.1 Minimum CC for automatic resolution estimationb\_values = \*Keep Zero Keep B values as supplied, or set all to zero. Only

applies if model is supplied.

- `translational_offsets` = \*None Any Allowed\_by\_symmetry Allow origin shifts that are compatible with the map or map coefficients. For cryo-EM data, use None for no offsets or Any if you want to allow translations. For Xray data, use None for no offsets or Allowed\_by\_symmetry to allow origin offsets compatible with the space group.
- `minimum_cc_for_auto_resolution` = 0.1 Minimum CC for automatic resolution estimation
- `b_values` = \*Keep Zero Keep B values as supplied, or set all to zero. Only applies if model is supplied.
- `input_filesmodel` = None `Modelmap_in_1` = None Map 1 `map_in_2` = None Map 2 `map_coeffs_1` = None Map coefficients 1 `map_coeffs_1_virtual` = None Used internally `map_coeffs_2_virtual` = None Used internally `map_coeffs_2` = None Map coefficients 2 `map_coeffs_labels_1` = None `map_coeffs_labels_2` = None
- `model` = None Model
- `map_in_1` = None Map 1
- `map_in_2` = None Map 2
- `map_coeffs_1` = None Map coefficients 1
- `map_coeffs_1_virtual` = None Used internally
- `map_coeffs_2_virtual` = None Used internally
- `map_coeffs_2` = None Map coefficients 2
- `map_coeffs_labels_1` = None
- `map_coeffs_labels_2` = None
- `controlclean_up` = True Clean up temp directories `verbose` = False Verbose output
- `clean_up` = True Clean up temp directories
- `verbose` = False Verbose output
- `directories` `temp_dir` = None Optional temporary directory `gui_output_dir` = None GUI use only - does not apply to command line version
- `temp_dir` = None Optional temporary directory
- `gui_output_dir` = None GUI use only - does not apply to command line version
- `map_model_ccresolution` = None Data (map) resolution `scattering_table` = wk1995 it1992 n\_gaussian neutron \*electron Scattering table (X-ray, neutron or electron) `atom_radius` = None Atom radius for masking. If undefined then calculated automatically `keep_map_calc` = False Keep model-calculated mapwrapping = None You can specify whether your maps wrap around outside boundaries `ignore_symmetry_conflicts` = False You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with `box_map` for example. `compute_cc_per_chain` = True Compute local model-map CC for each chain `cc_per_residue` = True Compute local model-map CC for each residue `cc_per_residue_group` = False Compute local model-map CC for each residue group `fsc` = True Compute FSC `cc_mask` = True `cc_volume` = True `cc_peaks` = True `cc_box` = True `cc_image` = False
- `resolution` = None Data (map) resolution
- `scattering_table` = wk1995 it1992 n\_gaussian neutron \*electron Scattering table (X-ray, neutron or electron)
- `atom_radius` = None Atom radius for masking. If undefined then calculated automatically

- `keep_map_calc = False` Keep model-calculated map
- `wrapping = None` You can specify whether your maps wrap around outside boundaries
- `ignore_symmetry_conflicts = False` You can ignore the symmetry information (CRYST1) from coordinate files. This may be necessary if your model has been placed in a box with `box_map` for example.
- `compute_cc_per_chain = True` Compute local model-map CC for each chain
- `cc_per_residue = True` Compute local model-map CC for each residue
- `cc_per_residue_group = False` Compute local model-map CC for each residue group
- `fsc = True` Compute FSC
- `cc_mask = True`
- `cc_volume = True`
- `cc_peaks = True`
- `cc_box = True`
- `cc_image = False`
- `outputfile_name_prefix = None` Optional prefix (including path) for output files. Default is name of the program and run number
- `file_name_prefix = None` Optional prefix (including path) for output files. Default is name of the program and run number
- GUI-specific parameters, not used on command line
- `result_file = None`
- `job_title = None` Job title in PHENIX GUI, not used on command line

# Mapping one PDB file onto another using space-group symmetry with phenix.map\_to\_object

---

map\_to\_object.html

## Author(s)

- phenix.map\_to\_object: Tom Terwilliger

## Purpose

phenix.map\_to\_object is a command line tool for applying a rotation and translation consistent with space-group symmetry to a PDB file in order to bring its atoms close to those in a second PDB file.

## Usage

### How phenix.map\_to\_object works:

phenix.map\_to\_object searches over each equivalent position in the unit cell and neighboring unit cells to find the one that places the moving\_pdb atoms closest to those in fixed\_pdb.

You can choose to minimize the distance between the center of mass of the PDB files, or you can minimize the distance between the closest atoms, or you can maximize the number of close contacts.

## Examples

### Standard run of phenix.map\_to\_object:

Running phenix.map\_to\_object is easy. You can just type:

```
phenix.map_to_object fixed_pdb=my_target.pdb moving_pdb=my_ligand.pdb
```

and phenix.map\_to\_object will move my\_ligand.pdb as close as it can to my\_target.pdb.

### Run of phenix.map\_to\_object specifying center of mass of moving PDB is to be close to any atom of fixed PDB:

By default phenix.map\_to\_object will move the center of mass of moving\_pdb as close as possible to any atom in fixed\_pdb. You could specify this explicitly with:

```
phenix.map_to_object fixed_pdb=my_target.pdb moving_pdb=my_ligand.pdb \  
use_moving_center_of_mass=True use_fixed_center_of_mass=False
```

### Run of phenix.map\_to\_object specifying center of mass of moving PDB is to have maximum number of contacts with atoms of fixed PDB:

If you wanted instead to maximize the number of close contacts under 5 Å between the center of mass of my\_ligand.pdb and any atom in my\_target.pdb, you could type:

```
phenix.map_to_object fixed_pdb=my_target.pdb moving_pdb=my_ligand.pdb \  
use_moving_center_of_mass=True use_fixed_center_of_mass=False \  
use_contact_order=True contact_dist=5.
```

## Run of phenix.map\_to\_object searching over additional unit cells

If you wanted additionally to search over more than just the adjacent unit cells to find the best contacts, you could add `extra_cells_to_search` with a value bigger than 1:

```
phenix.map_to_object fixed_pdb=my_target.pdb moving_pdb=my_ligand.pdb \
  use_moving_center_of_mass=True use_fixed_center_of_mass=False \
  use_contact_order=True contact_dist=5. \
  extra_cells_to_search=2
```

## Possible Problems

### Specific limitations and problems:

## Literature

## Additional information

### List of all available keywords

`map_to_object`  
`moving_pdb` = None PDB file with coordinates to move near `fixed_pdb` using SG symmetry  
`fixed_pdb` = None PDB file to move `moving_pdb` close to using SG symmetry  
`output_pdb` = None Name of output (moved) PDB file  
`use_moving_center_of_mass` = True You can choose to just move the center of mass of the moving PDB close to the fixed PDB (as opposed to finding the operator that puts an atom of the moving PDB closest to an atom in the fixed PDB)  
`use_fixed_center_of_mass` = False You can choose to just move the moving PDB close to the center of mass of the fixed PDB (as opposed to finding the operator that puts the moving PDB closest to any atom in the fixed PDB)  
`use_contact_order` = True You can choose to maximize the number of atoms that are within `contact_dist` (default=6) Å of an atom in the other structure.  
`map_by_chains` = False You can choose to move each chain individually. This is useful for taking a fragmentary model and mapping it on to a fixed model.  
`one_atom_per_residue` = None You can choose to use only one atom per residue for the search. Default is True if number of atoms is > 200.  
`contact_dist` = 6. Atoms separated by `contact_dist` or less are considered to be in contact.  
`extra_cells_to_search` = 1 You can specify how many unit cells beyond the central one to search in each direction (default=1, search -1 0 and 1 in each direction).  
`verbose` = False Verbose output.  
`raise_sorry` = False Raise sorry if problems.  
`debug` = False Debugging output.  
`dry_run` = False Just read in and check parameter names.  
`output_file` = target\_output\_format = \*None pdb mmCIF Desired output format (if possible). Choices are None (try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.

- `map_to_object`  
`moving_pdb` = None PDB file with coordinates to move near `fixed_pdb` using SG symmetry  
`fixed_pdb` = None PDB file to move `moving_pdb` close to using SG symmetry  
`output_pdb` = None Name of output (moved) PDB file  
`use_moving_center_of_mass` = True You can choose to just move the center of mass of the moving PDB close to the fixed PDB (as opposed to finding the operator that puts an atom of the moving PDB closest to an atom in the fixed PDB)  
`use_fixed_center_of_mass` = False You can choose to just move the moving PDB close to the center of mass of the fixed PDB (as opposed to finding the operator that puts the moving PDB closest to any atom in the fixed PDB)  
`use_contact_order` = True You can choose to maximize the number of atoms that are within `contact_dist` (default=6) Å of an atom in the other structure.  
`map_by_chains` = False You can choose to move each chain individually. This is useful for taking a fragmentary model and mapping it on to a fixed model.  
`one_atom_per_residue` = None You can choose to use only one atom per residue for the search. Default is True if number of atoms is > 200.  
`contact_dist` = 6. Atoms separated by `contact_dist` or less are

considered to be in contact `extra_cells_to_search = 1` You can specify how many unit cells beyond the central one to search in each direction (default=1, search -1 0 and 1 in each direction) `verbose = False` Verbose output `raise_sorry = False` Raise sorry if problems `debug = False` Debugging output `dry_run = False` Just read in and check parameter names `output_filetarget_output_format = *None pdb mmCIF` Desired output format (if possible). Choices are None ( try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.

- `moving_pdb = None` PDB file with coordinates to move near `fixed_pdb` using SG symmetry
- `fixed_pdb = None` PDB file to move `moving_pdb` close to using SG symmetry
- `output_pdb = None` Name of output (moved) PDB file
- `use_moving_center_of_mass = True` You can choose to just move the center of mass of the moving PDB close to the fixed PDB (as opposed to finding the operator that puts an atom of the moving PDB closest to an atom in the fixed PDB)
- `use_fixed_center_of_mass = False` You can choose to just move the moving PDB close to the center of mass of the fixed PDB (as opposed to finding the operator that puts the moving PDB closest to any atom in the fixed PDB)
- `use_contact_order = True` You can choose to maximize the number of atoms that are within `contact_dist` (default=6 ) A of an atom in the other structure.
- `map_by_chains = False` You can choose to move each chain individually. This is useful for taking a fragmentary model and mapping it on to a fixed model
- `one_atom_per_residue = None` You can choose to use only one atom per residue for the search. Default is True if number of atoms is > 200
- `contact_dist = 6`. Atoms separated by `contact_dist` or less are considered to be in contact
- `extra_cells_to_search = 1` You can specify how many unit cells beyond the central one to search in each direction (default=1, search -1 0 and 1 in each direction)
- `verbose = False` Verbose output
- `raise_sorry = False` Raise sorry if problems
- `debug = False` Debugging output
- `dry_run = False` Just read in and check parameter names
- `output_filetarget_output_format = *None pdb mmCIF` Desired output format (if possible). Choices are None ( try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.
- `target_output_format = *None pdb mmCIF` Desired output format (if possible). Choices are None ( try to use input format), pdb, mmCIF. If output model does not fit in pdb format, mmCIF will be used. Default is pdb.

# phenix.map\_to\_structure\_factors: convert map into structure factors

---

map\_to\_structure\_factors.html

## Description

The program reads a CCP4 (MRC) formatted map and converts it into structure factors. Output: mtz file containing Fourier transform of input map: complex array (F, PHIF) of given (d\_min) resolution Array of amplitudes F (F\_ampl) and phases as HL coefficients

The program reads a CCP4 (MRC) formatted map and converts it into structure factors.

Output: mtz file containing

- Fourier transform of input map: complex array (F, PHIF) of given (d\_min) resolution
- Array of amplitudes F (F\_ampl) and phases as HL coefficients

## Notes

If the map has an origin that is not (0,0,0), by default that origin is preserved in the output mtz file (if you create a map from the output map coefficients, that map will superimpose on the input map). You can adjust the origin of the output map with the keywords: keep\_origin=False: this will shift the origin to (0,0,0) output\_origin\_grid\_units=100,100,100: this will shift the origin to (100,100,100)

If the map has an origin that is not (0,0,0), by default that origin is preserved in the output mtz file (if you create a map from the output map coefficients, that map will superimpose on the input map).

You can adjust the origin of the output map with the keywords:

- keep\_origin=False: this will shift the origin to (0,0,0)
- output\_origin\_grid\_units=100,100,100: this will shift the origin to (100,100,100)

output\_origin\_grid\_units=100,100,100: this will shift the origin to (100,100,100)

## Contact author

For questions, bug reports, feature requests: Pavel Afonine (PAfonine@lbl.gov)

## Command line usage examples:

```
% phenix.map_to_structure_factors map.ccp4 d_min=3
```

## Current limitations:

X-plor formatted map files are not supported.

## List of all available keywords

output\_file\_name = map\_to\_structure\_factors.mtz d\_min = None Resolution of output structure factors. Default is based on the gridding of the map and can lead to map coefficients that are at much higher resolution than the map.resolution\_factor = 1./3 Scale factor to guess resolution of output structure factors. A scale factor of 0.5 gives the highest-resolution data allowed by the map. Usual is 0.33 or 0.25.scale\_max = 99999 Maximum value of output map coefficients amplitudes. If None use volume scalingk\_blur = 1 Scale applied to HL coefficients. The HL coefficients are arbitrary as no error information is available. The HL

coefficients will have values of `k_blur` at low resolution, falling off with an effective B-value of `b_blur` at higher resolution. `b_blur = 100` Blurring applied to HL coefficients. The HL coefficients are arbitrary as no error information is available. The HL coefficients will have values of `k_blur` at low resolution, falling off with an effective B-value of `b_blur` at higher resolution. `box = False` You can choose to generate a full box of map coefficients based on the gridding of the map. Default is to generate map coefficients to a specific resolution. `keep_origin = True` Default (`keep_origin=True`) is to set the origin for the output map coefficients to be the same as the input map. A map calculated from the output map coefficients will superimpose on the input map. If `keep_origin=False` then the new origin will be at (0,0,0). Note: `keep_origin=True` is only available if the map covers an entire unit cell. It will be automatically set to `False` if less than a full unit cell is available. `output_origin_grid_units = None` You can set the origin of the output map (in grid units) `map_file = None` Input map file `job_title = None` Job title in PHENIX GUI, not used on command line `guiGUI-specific parameter required for output directory` `output_dir = None`

- `output_file_name = map_to_structure_factors.mtz`
- `d_min = None` Resolution of output structure factors. Default is based on the gridding of the map and can lead to map coefficients that are at much higher resolution than the map.
- `resolution_factor = 1./3` Scale factor to guess resolution of output structure factors. A scale factor of 0.5 gives the highest-resolution data allowed by the map. Usual is 0.33 or 0.25.
- `scale_max = 99999` Maximum value of output map coefficients amplitudes. If `None` use volume scaling
- `k_blur = 1` Scale applied to HL coefficients. The HL coefficients are arbitrary as no error information is available. The HL coefficients will have values of `k_blur` at low resolution, falling off with an effective B-value of `b_blur` at higher resolution.
- `b_blur = 100` Blurring applied to HL coefficients. The HL coefficients are arbitrary as no error information is available. The HL coefficients will have values of `k_blur` at low resolution, falling off with an effective B-value of `b_blur` at higher resolution.
- `box = False` You can choose to generate a full box of map coefficients based on the gridding of the map. Default is to generate map coefficients to a specific resolution
- `keep_origin = True` Default (`keep_origin=True`) is to set the origin for the output map coefficients to be the same as the input map. A map calculated from the output map coefficients will superimpose on the input map. If `keep_origin=False` then the new origin will be at (0,0,0). Note: `keep_origin=True` is only available if the map covers an entire unit cell. It will be automatically set to `False` if less than a full unit cell is available.
- `output_origin_grid_units = None` You can set the origin of the output map (in grid units)
- `map_file = None` Input map file
- `job_title = None` Job title in PHENIX GUI, not used on command line
- `guiGUI-specific parameter required for output directory` `output_dir = None`
- `output_dir = None`

## <string>

---

map\_value\_at\_point.html

phenix.map\_value\_at\_point: tool to compute map value at a given point.

Inputs:

MTZ file containing Fourier map coefficients or real-space map; coordinates of points at which the map value will be computed can be defined in two ways: either providing a triplet of xyz coordinates using 'point' keyword, or giving a PDB file in which case the atomic coordinates will serve as points; optionally the 'grid\_step' can be specified. It defines the map calculation grid step in case Fourier map is provided optionally, type of map scaling can be defined: sigma or volume scaled (sigma is the default). Only used if Fourier map is provided.

- MTZ file containing Fourier map coefficients or real-space map;
- coordinates of points at which the map value will be computed can be defined in two ways: either providing a triplet of xyz coordinates using 'point' keyword, or giving a PDB file in which case the atomic coordinates will serve as points;
- optionally the 'grid\_step' can be specified. It defines the map calculation grid step in case Fourier map is provided
- optionally, type of map scaling can be defined: sigma or volume scaled (sigma is the default). Only used if Fourier map is provided.

Usage examples:

```
phenix.map_value_at_point map_coeffs.mtz point="0.5 1.2 3" point="0.53 3.3 2.5"
```

```
phenix.map_value_at_point map_coeffs.mtz model.pdb miller_array.labels.name="FMODEL"
```

```
phenix.map_value_at_point map.mrc model.pdb point="1 2 3" IMPORTANT: point must be a triplet of Cartesian coordinates (not fractional)!
```

- phenix.map\_value\_at\_point map\_coeffs.mtz point="0.5 1.2 3" point="0.53 3.3 2.5"
- phenix.map\_value\_at\_point map\_coeffs.mtz model.pdb miller\_array.labels.name="FMODEL"
- phenix.map\_value\_at\_point map.mrc model.pdb point="1 2 3"

IMPORTANT: point must be a triplet of Cartesian coordinates (not fractional)!

Output:

list of map values at specified points.

# Creating maps and map coefficients with phenix.maps

---

maps.html

## Contents

- How to run
- Remarks
- FFT-ing map coefficients
- Other map types

phenix.maps is a simple command-line tool for calculating map coefficients from a model and reflections file, with optional FFT to generate an actual map file. The program is largely self-documenting; this section essentially reproduces the command-line output of the program when run without arguments. As of PHENIX version 1.6.4, much of the parameter syntax is identical in phenix.maps and phenix.refine. There is also a GUI for phenix.maps, with separate documentation, but most of the information below applies to both the command-line and graphical versions.

## How to run

Run phenix.maps without any arguments: just type phenix.maps in the command line and hit Enter. This will create a parameter file called maps.params, which can be renamed if desired. Edit maps.params file to specify input/output file names, data labels and the desired maps. It is possible to request as many maps as desired. By default, the file maps.params specifies 5 maps to be created: 2mFo-DFc, 2mFo-DFc with missing Fobs filled with DFcalc, mFo-DFc and anomalous difference maps will be output in MTZ format, and one 2mFo-DFc map will be output in X-plor formatted file. Run this command to compute requested maps:

```
phenix.maps maps.params
```

- Run phenix.maps without any arguments: just type phenix.maps in the command line and hit Enter. This will create a parameter file called maps.params, which can be renamed if desired.
- Edit maps.params file to specify input/output file names, data labels and the desired maps. It is possible to request as many maps as desired. By default, the file maps.params specifies 5 maps to be created: 2mFo-DFc, 2mFo-DFc with missing Fobs filled with DFcalc, mFo-DFc and anomalous difference maps will be output in MTZ format, and one 2mFo-DFc map will be output in X-plor formatted file.
- Run this command to compute requested maps: phenix.maps maps.params

## Remarks

The scope of parameters 'map\_coefficients' defines the map that will be output as Fourier map coefficients. The scope of parameters 'map' defines the map that will be output as X-plor or CCP4 formatted map. To create several maps: duplicate either 'map\_coefficients' or 'map' or both scopes of parameters as many times as many maps is desired. Then edit each of them to define the maps. A map is defined by specifying a map type using 'map\_type' keyword available within each scope of parameters: 'map\_coefficients' or 'map'. The general supported format for 'map\_type' is: [p][m]Fo+[q][D]Fc[filled]. For example: 2Fo-Fc, 2mFobs-DFcalc, 3Fobs-2Fmodel, Fo-Fc, mFobs-DFcalc, anom. The 'map\_type' parser will automatically recognize which map is requested. The program creates an individual map file for each X-plor/CCP4 map defined in the parameters, but it creates only one MTZ formatted file with all Fourier map coefficients in it. The X-plor formatted map can be computed in the entire unit cell or around selected atoms only. Twinning (if detected) will be accounted for automatically. This can be disabled by using "skip\_twin\_detection=True" keyword. All arrays used in map

calculation, for example: Fobs, Fmodel, Fcalc, Fmask, m, D, etc., can be output into a CNS or MTZ formatted reflection file. For those who likes to experiment: bulk solvent correction and anisotropic scaling can be turned off, the data can be filtered by sigma and resolution.

- The scope of parameters 'map\_coefficients' defines the map that will be output as Fourier map coefficients. The scope of parameters 'map' defines the map that will be output as X-plor or CCP4 formatted map.
- To create several maps: duplicate either 'map\_coefficients' or 'map' or both scopes of parameters as many times as many maps is desired. Then edit each of them to define the maps.
- A map is defined by specifying a map type using 'map\_type' keyword available within each scope of parameters: 'map\_coefficients' or 'map'. The general supported format for 'map\_type' is: [p][m]Fo+[q][D]Fc[filled]. For example: 2Fo-Fc, 2mFobs-DFcalc, 3Fobs-2Fmodel, Fo-Fc, mfobs-Dfcalc, anom. The 'map\_type' parser will automatically recognize which map is requested.
- The program creates an individual map file for each X-plor/CCP4 map defined in the parameters, but it creates only one MTZ formatted file with all Fourier map coefficients in it.
- The X-plor formatted map can be computed in the entire unit cell or around selected atoms only.
- Twinning (if detected) will be accounted for automatically. This can be disabled by using "skip\_twin\_detection=True" keyword.
- All arrays used in map calculation, for example: Fobs, Fmodel, Fcalc, Fmask, m, D, etc., can be output into a CNS or MTZ formatted reflection file.
- For those who likes to experiment: bulk solvent correction and anisotropic scaling can be turned off, the data can be filtered by sigma and resolution.

## FFT-ing map coefficients

If you already have map coefficients in an MTZ file and just want to run an FFT to produce X-plor or CCP4 map files, use phenix.mtz2map. In most cases this can be run automatically by passing the name of the MTZ file. MTZ files generated by phenix.maps, phenix.refine, SOLVE/RESOLVE, and REFMAC are supported, and additional combinations of map coefficients are possible if defined by the user. By default, output files will take the base name of the MTZ file, with the appropriate map type appended if it can be guessed from the column labels, otherwise a unique integer, followed by the file type extension (.ccp4 or .xplor).

## Other map types

The functionality in phenix.maps is intentionally kept simple, but more complex map calculations may be performed in other programs:

For isomorphous difference maps (Fobs-Fobs), use phenix.fobs\_minus\_fobs\_map (documentation to be added). For omit maps, including composite omit maps and iterative build omit maps, use the AutoBuild wizard. You can also generate a simple simulated-annealing omit map by deleting the relevant part of the molecule and running phenix.refine with simulated\_annealing=True. (This is essentially the same was what is done by AutoBuild, however.)

- For isomorphous difference maps (Fobs-Fobs), use phenix.fobs\_minus\_fobs\_map (documentation to be added).
- For omit maps, including composite omit maps and iterative build omit maps, use the AutoBuild wizard.
- You can also generate a simple simulated-annealing omit map by deleting the relevant part of the molecule and running phenix.refine with simulated\_annealing=True. (This is essentially the same was

what is done by AutoBuild, however.)